

EDGE SPREAD OF HEMLOCK DWARF MISTLETOE AND IMPLICATIONS FOR THE GROUP
RETENTION SILVICULTURAL SYSTEM

by

HANNO SOUTHAM

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
DEGREE OF

MASTER OF SCIENCE

in

THE FACULTY OF GRADUATE AND POSTDOCTORAL STUDIES

(Forestry)

THE UNIVERSITY OF BRITISH COLUMBIA

(Vancouver)

April/May 2025

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

Edge spread of hemlock dwarf mistletoe and implications for the group retention silvicultural system

submitted by Hanno Southam in partial fulfilment of the requirements for

the degree of Master of Science

in Forestry

Examining Committee:

Dr. Richard Hamelin, Professor and Department Head, Forest and Conservation Sciences, UBC
Supervisor

Dr. Gregory Paradis, Assistant Professor, Forest Resources Management, UBC
Supervisory Committee Member

Dr. Ignacio Barbeito, Assistant Professor, Forest Resources Management, UBC
Supervisory Committee Member

Dr. Bianca Eskelson, Associate Professor, Forest Resources Management, UBC
Additional Examiner

Abstract

Variable retention silvicultural systems, defined by leaving mature trees in patches (group retention) or dispersed arrangements (dispersed retention), are used ~30% of harvested area in coastal British Columbia to balance timber and ecological objectives. Silvicultural systems (how a stand is harvested, regenerated and tended) affect forest pathogens, which creates complex decisions where foresters match associated impacts (positive and negative) of pathogens and silvicultural systems to their objectives. This study measured infection patterns of hemlock dwarf mistletoe (**HDM**; *Arceuthobium tsugense* (Rosend.) G.N. Jones subsp. *tsugense*), a native hemi-parasitic plant on western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), to infer additional infection from mature tree patches in group retention relative to clearcut systems. Infection patterns were measured at 11 sites along 55 m edges separating mature forest harbouring HDM (>100 years old) from a regenerating clearcut (23–45 years old). HDM infection was assessed with a six-class system rating system (**DMR**) that ranges from 0 (uninfected)–6 (>50% all branches infected). Mature forests (the infection source) had high incidence (95%) and variable average DMR (range = 1.4–5.7) and infected basal area (range = 18–106 m²/ha). Spread into regenerating forests was variable across infection boundary (3–32 m from edge), incidence (range, trees ≤15 m from edge = 0–100%) and average DMR (range, trees ≤15 m from edge = 0–3.0). Model predictions for a median site indicated light and moderate infection severities were most probable <10 m from the edge; beyond 15 m healthy trees were most probable and only light severity infection plausible, but unlikely (<20 %). HDM development to severities associated with growth impacts (DMR ≥3) over rotations of ~80 years is anticipated to be restricted to ≤15 m from the edge. Inferred to the stand-level and informed by the regional timber supply where hemlock is a lower-valued species, the potential impact to revenue from HDM in group retention is small. This supports the tentative conclusion that guidelines discouraging group retention where HDM occurs could be updated. Future research should predict stand-level infection, better characterize hemlock's role in timber supply and factor HDM ecological functions into decision-making.

Lay summary

Silvicultural systems are methods for harvesting, regenerating and tending forest stands. These systems affect forest pathogens—organisms that negatively influence forest plants and trees but play key ecological roles. Professional foresters in British Columbia rely on evidenced-based guidelines for silvicultural systems to make decisions that balance timber and ecological objectives. This research studied spread patterns of hemlock dwarf mistletoe, a parasitic plant that infects the timber species western hemlock, from mature stand edges into adjacent regenerating stands. These patterns can be used to make inferences about anticipated timber impacts in group retention, a silvicultural system that leaves patches of mature forest during harvesting to replicate natural disturbance. The results indicate infection levels associated with growth impacts are restricted to near the mature forest edge and have limited potential to impact revenue. This is a first datapoint for reassessing guidelines that discourage group retention in stands with hemlock dwarf mistletoe.

Preface

This project was done in collaboration with the British Columbia Ministry of Forests (**BC MoF**), who provided partial funding. The idea for the research was identified by Stefan Zeglen (retired Coast Region Forest Pathologist, BC MoF) and David Rusch (current Coast Region Forest Pathologist, BC MoF). The edge-infection research sites were established as part of the BC MoF's Experimental Project network (EP.1907). I led the research design, fieldwork, data analysis (except the technoeconomic analysis) and writing. Dr. Richard Hamelin oversaw all my work. My supervisory committee members (Dr. Gregory Paradis and Dr. Ignacio Barbeito) and David Rusch provided guidance on survey methods, site selection and data analysis and feedback on drafts of this thesis; Dr. Gregory Paradis filled an open space on my committee beginning in Fall 2024 and therefore could only contribute to the later stages of the project. Noel Hancocks (Undergraduate Research Assistant, Hamelin Lab) supported site identification and fieldwork throughout the summer of 2023. The technoeconomic analysis (sections 2.2 and 3.3) was performed by Yunhao (Davis) Xu, with guidance from Dr. Gregory Paradis. No part of this thesis has been published.

I used GenAI (ChatGPT) to draft code for analysis and to answer questions about concepts in data analysis and statistics. In analysis, I used it to accomplish coding tasks, not direct my approach. I verified all answers provided by GenAI relating to questions about data analysis and statistics with reliable sources (textbooks or journal articles).

Formatting follows the Chicago Manual of Style 18th edition. References follow the Chicago author-date system.

Table of contents

Abstract.....	iii
Lay summary	iv
Preface	v
Table of contents.....	vi
List of tables	viii
List of figures.....	ix
List of abbreviations	xi
Acknowledgements.....	xiv
Dedication.....	xv
Land acknowledgement.....	xvi
Chapter 1: Introduction.....	1
1.1. Research problem and gap.....	5
1.2. Research objective, questions and hypotheses	7
Chapter 2: Methods.....	9
2.1. Edge infection field study.....	9
2.1.1. Study sites.....	9
2.1.2. Field data collection.....	13
2.1.3. Data analysis.....	16
2.1.4. Seed load.....	18
2.2. Technoeconomic analysis of hemlock's role in timber supply	25
Chapter 3: Results.....	27
3.1. Site-level summaries of infection	27
3.2. Modelling results	34
3.3. Hemlock's contribution to timber supply	37
Chapter 4: Discussion	39
4.1. Edge infection patterns	39
4.2. Anticipated timber impacts at edge, stand and regional scales	43
4.3. Ecological functions of HDM and retention patches.....	46
4.4. Limitations	49

Chapter 5: Management implications and future research	52
5.1. Priority 1: Problem characterization with existing data	52
5.2. Priority 2: Continued research	53
Appendix.....	55
References.....	70

List of tables

Table 1. Characteristics of regenerating forest components.....	13
Table 2. Characteristics of mature forest components.....	13
Table 3. Dwarf mistletoe rating (DMR) system criteria for live, susceptible, trees.....	15
Table 4. Metrics used to describe HDM infection at the site level.....	17
Table 5. Summary of BC Ministry of Forests permanent sample plot dataset (PSP) used to fit height to live crown model.	20
Table 6. Summary of linear mixed effect model predicting height to live crown (m) from tree height (m) for hemlock trees in the mature component.	20
Table 7. Summary of crown volume and underlying tree dimensions, for hemlock trees in the mature component.....	21
Table 8. Interception factors for source tree-intervening tree pairs.....	23
Table 9. Summary of seed production and seed load proxies.	23
Table 10. Log prices by species used in technoeconomic analysis.	26
Table 11. Summary metrics of hemlock dwarf mistletoe infection in live hemlock trees in the mature forest component (infection source).	30
Table 12. Summary metrics of hemlock dwarf mistletoe infection in live hemlock trees in the regenerating component.....	33
Table 13. Summary of the modelling dataset.	34
Table 14. Summary of final model relating tree-level predictors to adapted dwarf mistletoe rating (DMR).	35
Table 15. Summary of timber supply and value creation potential (VCP) by major species in the two managed forest areas used in the case study.....	38
Table A1. Site selection criteria.....	55
Table A2. Climate and physical characteristics of the study sites.....	55
Table A3. Site history of regenerating forest components.	60
Table A4. Observers present at each study site.	60
Table A5. Model fit statistics for full suite of variable combinations tested during model building.	64
Table A6. Likelihood ratio test results of variable significance during model building.	65

List of figures

Figure 1. Overview of sites and survey methods.....	11
Figure 2. Location of study sites in southwestern, coastal, British Columbia, Canada.....	12
Figure 3. Relationship between height to live crown and tree height in hemlock trees in the BC Government Permanent Sample Plot data.	21
Figure 4. Summary of steps to calculate seed load proxy variable.	24
Figure 5. Stem maps showing hemlock dwarf mistletoe (HDM) infection at six representative sites.....	29
Figure 6. Metrics summarizing hemlock dwarf mistletoe infection in live hemlock trees in the dominant/codominant crown class at each site.....	30
Figure 7. Distribution of hemlock dwarf mistletoe (HDM) infection in hemlock trees by DBH in the mature forest components at each site.	31
Figure 8. Hemlock dwarf mistletoe (HDM) infection in hemlock trees in the regenerating forest component in 5.0 m intervals from the edge.	32
Figure 9. Mean contribution of each crown third to tree-level dwarf mistletoe rating (DMR) in regenerating forest component.	33
Figure 10. (A) Effect of increasing distance from the edge on predicted probabilities of dwarf mistletoe rating (DMR) classes for live dominant/codominant hemlock. (B) Seed load values observed in raw data with increasing distance from the edge.	36
Figure 11. Between site variation in predicted probabilities of dwarf mistletoe rating (DMR) classes for dominant/codominant hemlock.	37
Figure 12. Hypothetical clearcut and group retention cutblocks showing zones of edge influence where HDM impacts might occur.	45
Figure A1. Climate profiles of the study sites.	56
Figure A2. Monthly precipitation (mm) as snow and rain at each study site.	57
Figure A3. Difference between site precipitation and mean precipitation across sites for each month (mm).	57
Figure A4. Size class distribution and composition of regenerating forest components across the study sites.	58
Figure A5. Size class distribution and composition of mature forest components across the study sites...59	
Figure A6. Stem maps showing hemlock dwarf mistletoe (HDM) infection at five sites omitted in main text figure.....	61

Figure A7. Dwarf mistletoe rating (DMR) plotted against tree-level predictor variables.....	62
Figure A8. Pairs plot to assess collinearity between tree-level predictor variables.	63
Figure A9. Graphical test of proportional odds assumption for each predictor variable.	66
Figure A10. Predicted probabilities of each dwarf mistletoe rating (DMR) class from models with (PO) and without (PPO) proportional odds assumption on distance from the edge variable.....	67
Figure A11. (A) Effect of DBH on predicted probabilities of dwarf mistletoe rating (DMR) classes for trees at 7.5 m from the edge. (B) Comparison of DBH and distance from the edge effects on predicted probabilities of DMR classes for trees 20.0 to 50.0 m from the edge.	68
Figure A12. Random site effect estimates for each site.	69

List of abbreviations

AIC: Akaike Information Criterion

Ax: *Populus* hybrid

BA: basal area

Ba: amabilis fir

BC: British Columbia

BEC: biogeoclimatic zone

Bg: *Abies grandis*

BIC: Bayesian information criterion

CC: crown class, where D: dominant, C: codominant, I: intermediate and S: suppressed

CR: Campbell River

CV: crown volume

Cw: cedar

CWH: coastal western hemlock biogeoclimatic zone

BD: distance from tree bole

dAIC: difference in AIC

DBH: diameter at breast height

DD₅: degree days above 5 °C

DF: degrees of freedom

DI: dead infected

DIST: distance from mature forest edge

DMI: dwarf mistletoe index

DMIM: dwarf mistletoe impact model

DMR: dwarf mistletoe rating

Dr: *Alnus rubra*

DU: dead uninfected

ED: distance from edge of tree crown

Fd: Douglas-fir

F_{HT}: height factor

FNWL: First Nation Woodland Licence

FV: Fraser Valley

FVS: Forest Vegetation Simulator

LCW: largest crown width

LR: likelihood ratio test statistic

HDM: hemlock dwarf mistletoe

Hw: hemlock

HT: height

IBLC: infected below live crown

Int: interval

M: mature component

MAP: mean annual precipitation

MAT: mean annual temperature

N: sample size

n.d.: no date

PAR: precipitation as rain

PAS: precipitation as snow

PO: proportional odds

PH: Pacific Highway

PPO: partial proportional odds

PSP: permanent sample plot

Q: quartile, where Q1: 25th percentile and Q3: 75th percentile

R: regenerating component

Rad_{gs}: growing season radians

RESULTS: Reporting Silviculture and Land Status Tracking System

spp: species within a genus

SCF: Spel kúmtn Community Forest

SD: standard deviation

SE: standard error

SI: site index (for hemlock)

SL: seed load

SP: seed production

Ss: *Picea sitchensis*

TASS: Tree and Stand Simulator

Tmin: minimum temperature

U: utility log grade

VCP: value creation potential

VRI: Vegetation Resource Inventory

Acknowledgements

Thanks first to my supervisor, Richard Hamelin, who took me on and provided an opening to research in forest management, which has turned out to be my jam. He has been a consistent support and his willingness to work on problems with me made the project possible.

I am grateful to my committee members (Dr. Ignacio Barbeito and Dr. Gregory Paradis) and collaborator at the BC Ministry of Forests (David Rusch) for their honest, constructive comments and guidance. Thanks especially, for your help grounding the project in terms that matter to foresters—they are the audience this research is meant to be relevant too.

Thank you to the organizations that provided supporting funding: BC Ministry of Forests, Moss Rock Park Foundation, National Science and Engineering Research Council (NSERC), Namkoong Family Fellowship and the Faculty of Forestry. In parallel, thanks to the project partners who were willing to find and host research sites—Ionut Aron at Malcom Knapp Research Forest, Kelly Kitsch (Cameron) at the Mission Municipal Forest and Erik Holbek (Barkley Community Forest and surrounding tenure).

I owe thanks to numerous others who helped in various ways. Thanks to the people that collected data in often brushy fieldwork (Noel Hancocks, River Knockaert, Taylor Holt). Thanks to Liam Irwin and the Integrated Remote Sensing Studio for the lending and know-how of GPS equipment. Thanks to Don Robinson for helping think through the modelling approach. Thanks to Bridget Guo and Yunhao (Davis) Xu from the FRESH lab for the modelling they have done to support this project. Thanks to the rest of the Hamelin lab for their comradery and tolerating my rudimentary knowledge of genetics; in particular, thanks to Berni van der Meer for reading a draft of this thesis and Carol Ritland for keeping me on track.

And of course, thanks to my family and friends. Jeremy Greenberg, Angelina Kellas, Myles Lehman, Dunavan Morris-Janzen, Rachel Pekelney and Nat MacMillan, my mom, dad and siblings—thanks.

Dedication

For Dad.

Land acknowledgement

At UBC, I studied on the traditional, ancestral and unceded territory of the xʷməθkʷəy̓əm (Musqueam).

The field sites in my research are on located on the ancestral territories of the Toquaht, Nuchatlaht

(Ucluelet sites), Wei Wai Kum and We Wai Kai (Liḡʷiḡdaǰʷ people), K'ómoks, We Wai Kai, Homalco,

Kwakiutl (Campbell River sites), Leq'á:mel, Semá:th, Kwantlen, Sq'éwlets, Máthexwi, Katzie (Malcom

Knapp and Mission sites), Lilwat and St'át'imc (case study areas near Pemberton). I am grateful for the

learning I have been able to do on these territories and for what I have been able to learn about Indigenous

forest management.

Chapter 1: Introduction

Silvicultural systems specify how a stand is harvested, regenerated and tended to meet a set of objectives (BC Ministry of Forests 2003). Approaches to these systems that are based on replicating natural patterns of disturbance are one of the primary tools used in sustainable forest management (Montréal Process Working Group 2015; Kuuluvainen et al. 2021). These approaches are based on the logic that “by mimicking natural tree mortality patterns and forest structures at multiple scales in forest management, it is possible to maintain or restore key ecological structures, and hence sustain communities, processes and biodiversity, along with associated ecosystem services in managed forest landscapes” (Kuuluvainen et al. 2021, 3). Variable retention is one of these approaches. Its defining element is the retention of portions of mature forest throughout a managed area; these retention patches protect associated communities and ecological functions, and replicate patterns of forest age and structure in an ecosystem that result from its natural disturbance regime (Beese et al. 2019; Martínez Pastur, Vanha-Majamaa, and Franklin 2020). Use of variable retention has been shown to mitigate biodiversity impacts across multiple guilds of taxa, increase structural complexity and protect carbon stocks (Beese et al. 2019; Simard et al. 2021).

An emerging challenge of non-traditional silvicultural systems, like variable retention, is their interaction with forest pathogens. Silvicultural systems impact the incidence and severity of pathogens through effects on the forest components and processes that regulate pathogen introduction (e.g. the number and distribution of infection sources), spread (e.g. species diversity and local climate) and forest recovery post-infection (e.g. natural regeneration to replace mortality); however, our understanding of these effects is largely conceptual and focused studies of pathogen epidemiology in non-traditional silvicultural systems are needed (Roberts et al. 2020).

This study is based in coastal British Columbia (**BC**), Canada where variable retention emerged in the 1990s and was rooted in a recognition of the negative ecological impacts of industrial clearcut logging (Beese et al. 2019). In BC, variable retention silvicultural systems have specific criteria—more

than 50% of the regenerating trees in a harvested area must be within one tree length of a retained mature tree and the mature trees must be left in the stand for at least one full stand rotation (BC Ministry of Forests 2003). The retention trees can be dispersed across a site (dispersed retention) or in groups (group retention). Of the land area in BC (95 million ha), 58% (55 million ha) is forested, of which 44% (24 million ha) is productive forest where timber harvesting is allowed. Ninety-two percent (22 million ha) of this timber-allotted land base occurs on public land, administered by the BC government, through a system of tenures and agreements. In coastal BC, 29% (19 000 ha) of the area harvested (65 000 ha) between 2006–2017 on public lands used a variable retention silvicultural system, usually group retention (statistics from Beese et al. 2019). Most of the remaining harvesting used the clearcut with reserves system, which can resemble group retention in practice, but lacks the focus on non-timber objectives and related criteria on the distribution and leave-time of retention trees (BC Ministry of Forests 2003; Beese et al. 2019).

Dwarf mistletoes (*Arceuthobium* M. Bieb.) are a group of hemi-parasitic flowering plants, distributed across the northern hemisphere, that infect trees in the Pinaceae and Cupressaceae (Hawksworth and Wiens 1996). Hemlock dwarf mistletoe (**HDM**; *Arceuthobium tsugense* (Rosend.) G.N. Jones subsp. *tsugense*) is the most common and abundant species in coastal BC (for comprehensive review see Muir and Hennon 2007). Its range extends throughout the Pacific Northwest (California to Alaska), overlapping with its primary host western hemlock (*Tsuga heterophylla* (Raf.) Sarg.). Hemlock is an abundant, but typically lower-valued timber species in coastal BC (Klinka et al. 2000; BC Ministry of Forests 2024a). At start of the dwarf mistletoe lifecycle (Hawksworth and Wiens 1996), a seed lands on host foliage, germinates and radicles penetrate young (<6 years old) branches through physical pressure at needle bases or branch flaps. (Hunt, Owens, and Smith 1996). An endophytic system is formed inside the branch consisting of longitudinally oriented cortical strands and radially oriented sinkers; the sinkers become intermingled with host rays and take up nutrients via apoplastic pathways (Calvin and Wilson 1996, 11). External shoots are produced as the infection matures, which eventually bear either male or female flowers (dwarf mistletoes are dioecious), produce pollen or ovules, and finally

fruit. For HDM, the period between initial infection and fruiting is around 3–5 years (Muir and Hennon 2007).

Dispersal in dwarf mistletoes occurs primarily by explosive seed discharge over ranges of 0–15 m (Hawksworth 1959; Smith 1966; DeBruyn et al. 2015) and less frequently by seed attached to bird and small mammal vectors (Ostry, Nicholls, and French 1983; Nicholls, Egeland, and Hawksworth 1989). Seeds are covered in a sticky substance (viscin) and adhere to animals when they are in the canopies of infected trees, then later fall off or are removed by preening. No coevolved seed vectoring mechanisms have been documented with animal vectors, but some birds appear to have a habitat preference for open canopies that dwarf mistletoes contribute to generating (Ostry, Nicholls, and French 1983; Nicholls, Egeland, and Hawksworth 1989). These dispersal mechanisms give rise to a modular pattern of infection centres at stand and landscape scales, initiated by animal vectoring, then enlarged by explosive seed discharge (Alfaro et al. 1985; Shaw et al. 2005; Swanson, Shaw, and Marosi 2006).

Dwarf mistletoes are characterized by dual associations that are common in native forest pathogens—they play key ecological roles and cause negative timber impacts when infecting merchantable timber species (Lundquist, Goheen, and Goheen 2002; Griebel, Watson, and Pendall 2017). The ecological roles of HDM are poorly studied but include generating canopy structure and wildlife nesting platforms by inducing dense branching (also called witches' brooms) in its host tree (e.g. for the endangered marbled murrelet, Hamer et al. 2008) and contributing to the development of old-growth stand structure, by amplifying small scale mortality (one to a few trees at a time) (Muir and Hennon 2007). HDM causes tree-level timber impacts when many infections accumulate in a tree, reducing height and diameter growth (by ~ 20% with moderate infection and ~ 40% with severe infection) (Thomson, Smith, and Alfaro 1984; Thomson et al. 1985; Shaw, Huso, and Bruner 2008), increasing drought mortality (Bell, Pabst, and Shaw 2020) and causing deformities and stem breakage that reduce wood quality (Muir and Hennon 2007).

Dwarf mistletoes obtain most of their water, mineral nutrients and carbohydrates from their hosts (Hull and Leonard 1964) but their small size limits their impact as resource sinks. The disproportionate

impacts to host trees associated with HDM infection occurs through transformations of the host crown and changes to branch and leaf physiology. HDM infected branches carry less water (measured as specific conductivity) and hold foliage with reduced photosynthetic capacity and water use efficiency; water availability (measured as leaf specific conductivity) to this photosynthetically-inefficient foliage is maintained by reducing the amount of foliage supported by infected branches (Marias et al. 2014; Meinzer, Woodruff, and Shaw 2004). This adaptive, allometric response to maintain leaf-level water relations manifests at the tree-level in trees with fewer live branches, less foliage and remaining foliage located distal to HDM infections (Calkins, Shaw, and Lan 2021). The cumulative impact of reductions in total foliage and the photosynthetic capacity of remaining foliage on carbon accumulation explain the reductions to height and diameter growth in infected trees described above (Calkins, Shaw, and Lan 2021; Meinzer, Woodruff, and Shaw 2004).

At a site level, the primary determinant of HDM infection levels is the number, distribution and characteristics of infected source trees (Parmeter 1978; Muir and Hennon 2007). Large, infected dominant/codominant trees exert the most infection pressure. These trees can support many infections, are abundant sources of seed and are in a canopy position that maximize the potential area affected by seed dispersed from them. They are contrasted by small sources trees in suppressed and intermediate canopy positions that support fewer infections and have reduced dispersal potential. Infection levels are regulated by the secondary influence of numerous tree factors (species, tree size, resistance, vigour) (Smith, Wass, and Meagher 1993; Bickford, Kolb, and Geils 2005; Shaw et al. 2005; Mathiasen and Kenaley 2017) and site factors (age, size structure, productivity, density, light environment, host and non-host abundance, slope, climate, weather) (Smith 1973; Bloomberg and Smith 1982; Shaw and Weiss 2000; Barrett et al. 2012) that affect HDM's lifecycle, dispersal and infection success. These factors control how HDM spreads (defined as establishing new infections on previously healthy trees) from source trees within a site and intensifies (defined as increasing the number of infections in an infected tree), which determines the extent of its impact on timber resources and non-timber forest components and processes (Parmeter 1978; Muir and Hennon 2007).

1.1. Research problem and gap

The interaction of HDM and variable retention silvicultural systems is a longstanding forest management challenge (Muir and Hennon 2007). HDM infection accumulates within trees and stands over time and severe infection most often occurs in mature stands. The mature trees retained in retention silvicultural systems can harbour HDM and act as infection sources, resulting in increased levels of infection relative to even-aged, clearcut systems. The anticipated timber impacts from this interaction has led to management guidelines in BC that recommend against using retention systems in stands with HDM, except where non-timber values are of exceptional importance (Rusch et al. 2019). However, HDM infection and the actual impact to hemlock timber volume and revenue in retention systems have been almost unstudied. In tandem, potential impacts from these guidelines on other forest components and processes affected by HDM infection and retention trees or patches have received little research.

With few available stand-scale studies (Deal, Tappeiner, and Hennon 2002; Muir and Hennon 2007, 99–102), efforts to anticipate HDM infection under different silvicultural systems have relied on extrapolating results from fine-scale studies that have measured HDM incidence and severity around residual trees; (Smith 1977; Bloomberg, Smith, and Van Der Wereld 1980; Bloomberg and Smith 1982; Shaw III 1982; Shaw III and Hennon 1991; Trummer et al. 1998); residual trees refer to infected, overtopping, often unmerchantable hemlock left during harvesting. Muir and Hennon (2007) use these studies to summarize expected patterns of HDM spread and intensification from infected edges (of a cutblock boundary or retention patch) and the implications for group retention stands over standard rotations (~80 years). They suggest infection will become noticeable at ~20 years post-harvest and be concentrated within 10 m of infected edges. This corresponds to the area under the greatest influence of overstory, infected edge trees, which is constrained by the dispersal limit of explosive seed discharge (~15 m) (Smith 1966; 1973; 1977). Infection in trees beyond 10 m is predicted to be rare and low severity, based on slow rates of spread and conditions that can limit HDM intensification in even-aged portions of a stand (Richardson and van der Kamp 1972; Bloomberg and Smith 1982; Shaw and Weiss

2000). At the stand scale in group retention, they predict levels of infection will depend on opening size; 10 m radius openings (~0.03 ha) will be fully colonized and whereas larger openings (e.g. 1 ha, typical of group retention in BC; Beese et al. 2019) will have much lower infection, proportional to the area beyond 10 m.

HDM development is markedly slower in northern portions of its range (Alaska, northern coastal BC) than in southern portions (southern coastal BC, Washington Oregon) (Shaw III 1982; Shaw III and Hennon 1991; Trummer et al. 1998); this latitudinal effect drives differences in expected incidence and severity of trees within the 10 m zone of concentrated infection around residual trees (Muir and Hennon 2007). In southern regions, most of these trees ($\geq 50\%$) are expected to be infected and beginning to reach severities associated with growth impacts by mid-rotation (~40 years). In northern regions, this tree group is expected to have somewhat lower incidence and not reach severities associated with growth impacts in a rotation.

The only studies of HDM infection in non-clearcut silvicultural systems are set in Alaska and conform with the expected slow rate of HDM development there. Deal, Tappeiner and Hennon (2002) compared paired uncut and selective partial harvest stands that range from 9-95 years-since-harvest; no significant differences in average HDM severity were observed between uncut and harvested stands in either overstory or young trees. They interpret this to indicate that selective partial harvests maintain HDM infection at levels similar to unmanaged stands. A small study reported in Muir and Hennon (2007, 99–102) reconstructs HDM infection in similar 60-year-old selective partial harvest stands and 110-year-old wind disturbed stands that mimic selective partial harvests. It is the only study estimating HDM associated stand-level growth impacts in a non-clearcut system. Volume loss is estimated relative to a HDM free stand condition for a single timepoint based on severity ratings and is small (range 1-6%).

The small number of focused studies of HDM infection in non-clearcut silvicultural systems is a research gap with direct links to forest management decisions. No studies have examined group retention systems. The selective partial harvest systems studied in Alaska resemble dispersed retention, but the studies are the only of their kind and do not generalize to southern regions of HDM's range. Extrapolating

results from fine-scale studies of HDM infection around residual trees has a sound rationale in dispersed retention silvicultural systems but has a weaker argument in group retention systems, where HDM infection originates from edges of cutblock boundaries and retention patches. Infection patterns from these edges have not been studied and numerous potential differences could cause them to differ from those around single residual trees (e.g. edges likely contain multiple HDM infected source trees that cumulatively determine infection in regenerating trees).

To approach this research gap, we conducted a field study measuring HDM infection patterns from edge infection sources to generate high resolution data that could be inferred to group retention silvicultural systems. This primary research effort was supplemented with a case study investigating the importance of hemlock in the coastal BC timber supply; this case study provided context to interpret the potential for anticipated impacts to hemlock growth to translate to impacts on revenue. Enabling the use of non-traditional silvicultural systems requires research and planning tools that support the foresters using the systems (Lundquist, Goheen, and Goheen 2002). Robust estimates of HDM infection in group retention silvicultural systems would allow foresters to make intentional decisions that match the associated impacts (positive and negative) to management objectives.

1.2. Research objective, questions and hypotheses

Research objective

Measure HDM edge infection patterns to provide a foundation for inferring stand-level infection and associated timber impacts to western hemlock in stands managed with clearcut and group retention silvicultural systems.

Research question 1

What is the pattern of HDM incidence and severity in regenerating stands adjacent to infected mature forests 20–45 years post-harvest?

Hypotheses

1. HDM incidence and severity will decline exponentially with distance from the mature forest edge, with highest incidence and severity located within the seed dispersal distance of mature source trees (≤ 15 m from edge).
2. HDM incidence will be $>50\%$ in trees ≤ 15 m from the edge and decline rapidly to $<10\%$ in trees >15 m.
3. HDM severity will range from light-moderate in trees ≤ 15 m from the edge and be limited to light in trees >15 m from the edge.

Research question 2

What proportion of harvested volume and revenue does hemlock account for in two managed forest areas near Pemberton BC?

Hypothesis: Hemlock will comprise an important share of harvested volume and will plausibly make (1) moderate, (2) small or (3) negative contributions to revenue.

Chapter 2: Methods

2.1. Edge infection field study

2.1.1. Study sites

Each site consisted of a 55.0 m long portion of the edge of a 20–45-year-old clearcut harvested area (**regenerating component**) that had adjacent mature (>100 year-old) forest with severe HDM infection (**mature component**) (Figure 1). We considered the mature component the infection source and the regenerating component the stand being recolonized. Eleven sites were located across southwestern, coastal, BC (Figure 2), based on criteria for age, composition, and site history (Table A1). Hemlock had to be a leading species ($\geq 30\%$) in both the regenerating and mature component. Sites were located within larger clearcut areas to ensure the mature component was the only infection source. The sites were established as permanent monitoring plots, and we intentionally tried to cover a range of site and climate conditions to enable future studies targeted at specific variables and represent a variety of managed stands.

Sites were identified by project partners that manage an area of forest or through public databases when sites occurred on public land. Sites on public land were identified in a two-step process: (1) coarse scale identification of potential sites with Vegetation Resource Inventory (**VRI**) data (measurements projected to 1 January 2021; Forest Analysis and Inventory Branch 2024b) and (2) fine scale site screening with data from the Reporting Silviculture Updates and Land Status Tracking System (**RESULTS**) (BC Ministry of Forests 2023). The VRI Rank 1 layer is a geospatial forest inventory dataset that is photo-based and validated by ground sampling. RESULTS is the central repository that tracks silviculture activities carried out on public land in BC. Forest health data is often poor in RESULTS and in-person screening to confirm the presence of HDM was required to identify the final set of sites.

Descriptive climate data for each site was sourced from ClimateBC v7.50 for the Normal 1991–2020 period (Wang et al. 2016). Selected variables have a link to HDM biology (Barrett et al. 2012) or are

common descriptors. All sites are in the coastal western hemlock (CWH) biogeoclimatic zone (Pojar, Klinka, and Meidinger 1987; Pojar, Klinka, and Demarchi 1991) where the HDM hazard is rated high (Rusch et al. 2019). The sites are in three geographic clusters that explain most of the variation in the selected climate variables between the sites (Table A2; Figure A1). Mean annual temperature varies little among the sites (8.7–9.9 °C) but the mean annual precipitation (MAP) range is relatively large (1772–4007 mm). The Pacific Highway cluster is the wettest (mean MAP = 3774 mm), Campbell River the driest (mean MAP = 1959 mm) and Fraser Valley a middle value (mean MAP = 2320 mm). Precipitation differences are explained disproportionately by winter months (Figure A2 and A3).

A range of successional stages, size distributions and compositions are represented in the sites (Table 1 and 2; Figure A4 and A5). Site index data for hemlock was sourced from provincial site productivity layer (Forest Analysis and Inventory Branch 2021) and ranges from 17.2–32.6 m (Table A2). Dominant tree species included hemlock, Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco), western red cedar (*Thuja plicata* Donn ex D.Don) and amabilis fir (*Abies amabilis* (Douglas ex Loudon) J. Forbes). Minor species were *Abies lasiocarpa* (Hook.) Nutt., *Acer circinatum* Pursh, *Acer macrophyllum* Pursh, *Alnus rubra* Bong., *Betula papyrifera* Marshall and *Prunus* L. spp. Amabilis fir and Douglas-fir are occasional and rare hosts, respectively, of HDM (Mathiasen and Kenaley 2017). The variation in composition and density across the sites is due in part to the site histories that established the regenerating components (Table A3). Site history data was missing for four of the sites and is tracked relative to the larger harvested area the sites are embedded in, meaning that even recorded treatments may not have overlapped the specific area covered by a site if they only occurred in portions of the harvested area. Of the areas with data, five were brushed and three received juvenile spacing (precommercial thinning); details on spacing were often lacking, but Douglas-fir and cedar are typically kept over hemlock. Douglas-fir and cedar were the most commonly planted species. Hemlock is generally listed as a preferred or acceptable species in site plans; it naturally seeds into harvested areas but is rarely planted. The dominance of hemlock at our sites reflects the position of the sites at the edge of harvested areas, and thus close to a seed source (i.e. hemlock trees in the mature component) (Harris 1967).

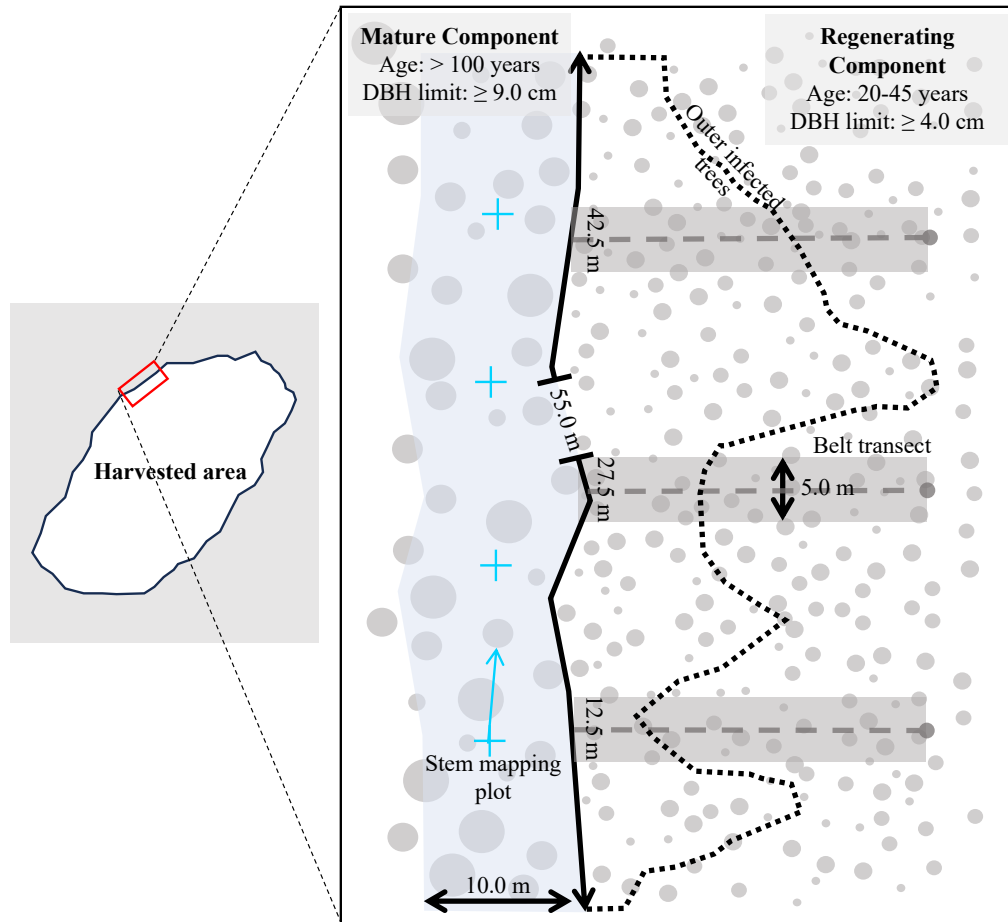


Figure 1. Overview of sites and survey methods. Each site consisted of a 55.0 m long portion of the edge of a clearcut harvested area. The harvested area must have been cut 20–45 years ago and is called the regenerating component. The adjacent mature forest is termed the mature component; it had to be >100 years-old and harbour severe hemlock dwarf mistletoe (HDM) infection. Hemlock had to be a leading species in both components. The mature component was represented by a 10.0 x 55.0 m survey area. The regenerating component was represented by 5.0 m wide, variable length transects that extended perpendicular to edge. Transect length was determined by the infected regenerating tree farthest from the edge. Trees were stem mapped, measured and rated for HDM infection.

Hemlock dwarf mistletoe edge spread research sites

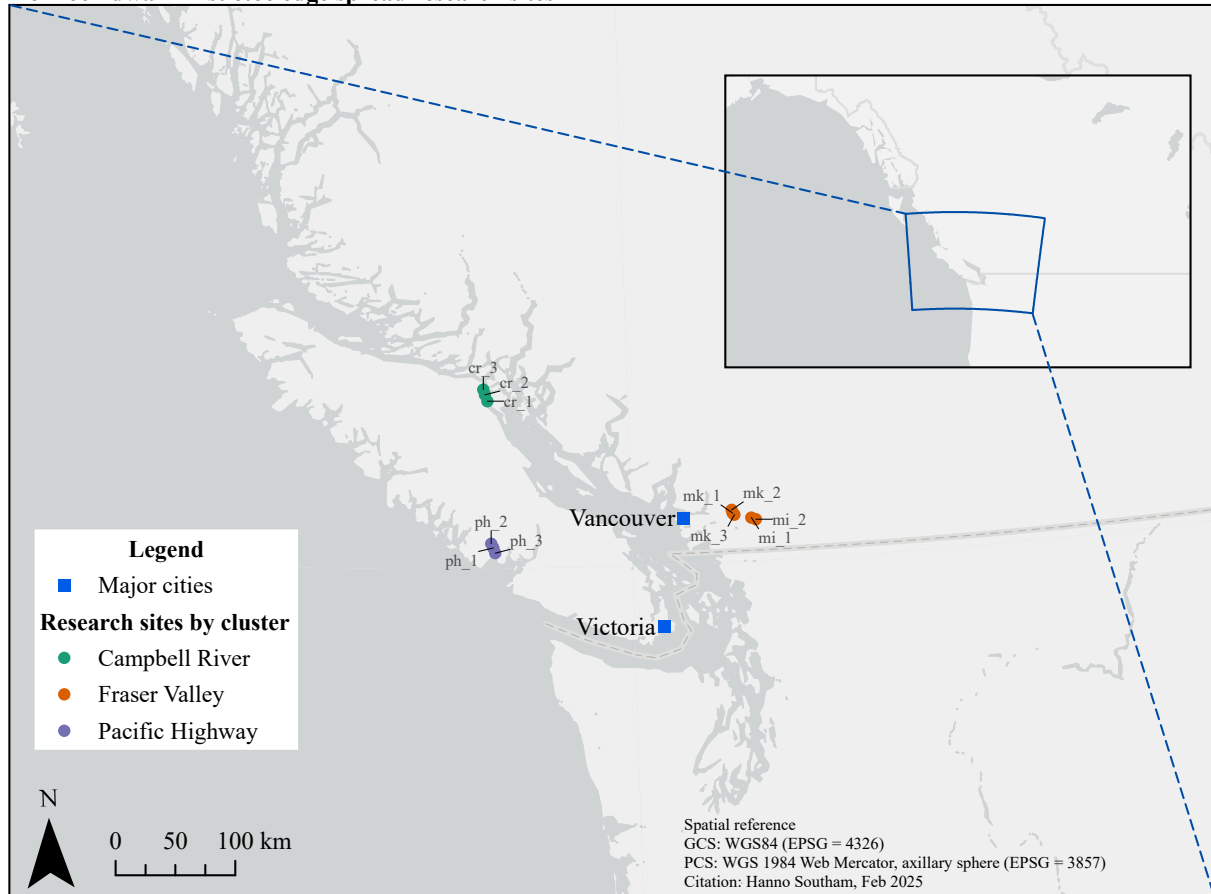


Figure 2. Location of study sites in southwestern, coastal, British Columbia, Canada. Point labels (e.g. cr_1) are site identifiers.

Table 1. Characteristics of regenerating forest components. Age estimate is years between harvest and measurement, which occurred in summer 2023 or 2024, depending on site. Height estimates are from a sub-sample of field measured trees (some exceptions, see footnote). Field measured trees include trees ≥ 4.0 cm DBH within one of three 5.0 m wide belt transects. Only data from 0.0–15.0 m on each transect, which was collected at every site, is used. Stem density and basal area are for all measured trees (live and dead). N_{HT} = number of height measurements, HT = height, SD = standard deviation, BA = basal area.

Site	Dominant and codominant hemlock						% trees hemlock	BA (m^2/ha)	% BA hemlock
	Age	N_{HT}	HT (m)	Mean DBH (cm)	SD DBH (cm)	Density (trees/ha)			
cr_1	33	12	22.3	22.3	4.6	1733	77	24	66
cr_2	38	12	26.7	30.1	2.9	1378	94	30	89
cr_3	37	14	21.5	21.3	5.3	1956	80	35	72
mi_1	32	9	22.4	23.1	6.2	2222	56	48	78
mi_2	29	-	17.0 ^v	16.4	3.4	3689	94	29	95
mk_1	33	5	17.6	15.6	4.5	3378	91	33	95
mk_2	45	7	23.7	20.5	4.7	2578	93	31	95
mk_3	23	8	15.3	15.8	5.6	2533	79	18	71
ph_1	40	12	19.6	22.9	5.0	5022	86	69	84
ph_2	35	8	20.2	20.5	3.9	3244	30	32	33
ph_3	33	5	14.4	17.7	5.8	2000	27	33	12

^v indicates value is from VRI data, projected to 1 January 2021

Table 2. Characteristics of mature forest components. Height estimates are from a sub-sample of field measured trees (some exceptions, see footnote). Field measured trees include trees ≥ 9.0 cm DBH within the 10.0 x 55.0 m survey area. Stem density and basal area are for all measured trees (live and dead). N_{HT} = number of height measurements, HT = height, SD = standard deviation, BA = basal area.

Site	Dominant and codominant hemlock						% trees hemlock	BA (m^2/ha)	% BA hemlock
	Age ^v	N_{HT}	HT (m)	Mean DBH (cm)	SD DBH (cm)	Density (trees/ha)			
cr_1	106	12	30.0	38.3	24.5	782	77	110	42
cr_2	146	12	31.9	44.9	18.9	927	67	140	44
cr_3	107	10	32.3	41.3	22.9	1000	69	145	41
mi_1	165	11	38.5	62.8	18.5	873	96	111	97
mi_2	99	-	34.9 ^v	36.4	13.2	1109	85	102	85
mk_1	87	3	42.0	51.2	38.8	1709	88	144	38
mk_2	157	7	38.9	50.3	11.6	764	76	115	45
mk_3	82	8	38.8	50.9	11.7	1145	86	93	60
ph_1	399	12	27.6	40.8	14.3	1564	99	116	92
ph_2	420	-	44.2 ^v	49.1	81.9	727	40	264	10
ph_3	260	5	23.7	34.5	10.8	745	49	55	38

^v indicates value is from VRI data. Height estimates are projected to 1 January 2021. Age estimates have been corrected to year surveyed.

2.1.2. Field data collection

Measurements occurred over two field seasons (June–August 2023, June–July 2024). Figure 1 shows the survey methods. At each site we mapped and assessed trees within 10.0 m of the 55.0 m long edge in the mature component and along three 5.0 m wide, variable length, belt transects that ran perpendicular to the edge into the regenerating component. Transect length was constant within a site and was determined by the distance to the infected tree farthest from the edge, or a minimum of 15.0 m. In

exceptional cases where transects ended in a location without trees (e.g. windthrow area), they were extended to capture the next hemlock. Transects were separated from each other by 15.0 m (locations on the edge line: 12.5 m, 27.5 m, 42.5 m). This spacing left 10.0 m of unsampled area between transects and between the ends of the 55.0 m edge line, resulting in a survey coverage of 27.3% (15.0/55.0 m).

We used the standard six-class dwarf mistletoe rating system (**DMR**) (Hawksworth 1977) with one adaptation to assess HDM infection in live susceptible trees (Table 3). With the standard system, the live crown of each live susceptible tree is divided into thirds and each third is given a rating of 0 (no infection), 1 ($>0\%$ and $\leq 50\%$ of primary branches infected) or 2 ($>50\%$ of primary branches infected); these ratings are recorded separately and then summed to give a rating out of 6 for the whole tree. We added an additional rating class to identify trees that had infections but only below the live crown (**IBLC**). This distinguishes these trees from trees with no infection. In each tree, we also recorded the presence and location of witches' brooms and stem infections, which indicate severe and old infections respectively. Dead trees were assessed for HDM on a presence/absence basis or recorded as unknown when insufficient tree material remained to assess. Trees in the regenerating component we suspected were residuals or infected advance regeneration from the pre-harvest stand were identified with characteristic signs (large tree size, severe HDM infection, brooms and/or stem infections). Ratings were completed by four observers over the course of data collection from the ground using binoculars (Table A4). At least two observers were present at each site. Ratings from observers involved in fewer sites were validated by more experienced observers. We frequently calibrated ratings between observers, with the goal of minimizing observer bias that has been documented with the DMR rating system (Shaw, Freeman, and Mathiasen 2000).

Table 3. Dwarf mistletoe rating (DMR) system criteria for live, susceptible, trees. The live crown was divided into thirds and each third was rated separately. Typically, the whole tree rating was the sum of the crown third ratings; the exception were trees that contain infections but only below live crown (IBLC).

Rating	Criteria
Crown third ratings	
0	No infected branches
1	>0% and ≤50% of primary branches infected
2	>50% of primary branches infected
Whole tree ratings	
0	No infections in tree
IBLC	Infections only located below live crown
1–6	Infections present in the live crown. Calculated as sum of crown third ratings

We measured trees ≥ 4.0 cm diameter at breast height (**DBH**) in the regenerating component and ≥ 9.0 cm DBH in the mature component, which aligns with cutoffs for 5.64 m radius (~ 0.01 ha) and 11.28 m radius (~ 0.04 ha) plots used in forest inventories in our region (BC Ministry of Forests 2024b). Other tree standard measurements included: species, status (live standing, live leaning, live fallen, recently dead, dead standing, snag), crown class (crown position relative to other trees; dominant, codominant, intermediate, suppressed), DBH, damage agents and pathological indicators (adapted from BC Ministry of Forests 2024b). Heights were taken for a subsample of dominant and codominant hemlock in each component at most sites; when a height sample was not taken because of time constraints on fieldwork, VRI data was substituted (1/11 regenerating components and 2/11 mature components) (Table 1). VRI data was projected to 2021, which is mismatched two or three growing seasons (depending on site) with field measurements (taken in 2023 or 2024).

Tree locations were recorded relative to differential GPS points and then converted to absolute positions in data processing. Differential correction was completed with base station data from the BC Active Control System (Province of BC 2024); horizontal precision of differential GPS points post-correction averaged 0.5 m (range = 0.0–2.2 m). Horizontal distance was used for plot layout to account for slope differences.

2.1.3. Data analysis

The length of transects in the regenerating component ranged from 15.0–51.0 m. To allow us to report infection levels for comparable areas between sites, we simulated data from the transect end point to 50.0 m. The stem density of each species-status-crown class combination was estimated using the field measured data from 5.0 m to the transect ends at each site; because the goal was to simulate trees extending toward the interior of the harvested area, data from 0.0–5.0 m was excluded to remove immediate edge effects. New trees were generated from these combinations and assigned a DBH from a normal distribution defined by the mean and standard deviation of a given combination at a site. Simulated hemlock trees were considered healthy ($DMR = 0$) because the transect end defined the outer infection limit.

HDM infection is summarized at the site level with the metrics in Table 4 for three tree groups:

1. **Regenerating component 0.0–15.0 m**, which contains only measured hemlock and is within the range (~ 15.0 m) of HDM seed dispersed by explosive discharge from trees in the mature component.
2. **Regenerating component 0.0–50.0 m**, which contains measured and simulated hemlock and extends beyond the range of HDM seed dispersed from trees in the mature component by explosive seed discharge.
3. **Mature component**, which contains all measured hemlock in the mature component boundary.

To calculate the average DMR and dwarf mistletoe index (**DMI**) metrics, trees in the IBLC DMR class were converted to $DMR = 0$, which aligns with the original definitions of the metrics (Geils and Mathiasen 1990). Secondary hosts (amabilis fir and Douglas-fir) are excluded. HDM infection was extremely rare in Douglas-fir, and we treated it as a non-host. Amabilis fir was only present at one site (ph_2) and may be an important contributor to HDM infection there; infection summary statistics are reported for it at that site separately. Similarly, suspected residual or infected advanced regeneration hemlock trees were excluded to focus on the tree cohort that originated post-harvest.

Table 4. Metrics used to describe HDM infection at the site level. Typically, metrics area summarized for the mature component and the two intervals (0.0–15.0 m and 0.0–50.0 m) in the regenerating component. Basal area infected and infection perimeter are exceptions (see footnotes).

Metric	Definition
Incidence	% live hemlock trees that are infected (DMR = IBLC or ≥ 1)
DMR ≥ 3 incidence	% live hemlock trees with infection severities associated with growth impacts (DMR ≥ 3)
Brooming incidence	% of live hemlock trees with witches' brooms
Stem infection incidence	% of live hemlock trees with HDM infections on their bole
Average DMR	Mean DMR of all live hemlock at a site. Trees with DMR = IBLC were converted to DMR = 0
Dwarf Mistletoe Index (DMI)	Mean DMR of all live hemlock at a site with DMR ≥ 1 . Trees with DMR = IBLC were excluded
Basal area infected (m^2/ha) ¹	Basal area of infected live and dead hemlock trees
Infection perimeter (m) ²	Horizontal distance from the edge that captures 90% of live infected hemlock

¹Only presented for the mature component because differences in age and site histories in the regenerating component confound comparisons between sites

²Only presented for the regenerating component because it represents spread from the mature component

To statistically describe HDM infection in the regenerating component with increasing distance from edge, tree DMR was modelled as a function of tree-level variables and a random site effect with cumulative logit random intercept models (Agresti 2010). Our modelling objective was inferential—to describe the effect of the predictor variables (particularly distance from the edge) on HDM infection. Live hemlock trees over the 0.0–50.0 m interval (including measured and simulated trees) were used as the modelling dataset. Dead hemlock trees were excluded because mortality at the age class in our data is not caused by HDM infection and live trees are future merchantable trees. DMR was collapsed to three classes, 0 (healthy), IBLC–2 (light infection) and 3–5 (moderate–severe infection), to treat model convergence issues associated with data scarcity; these groups correspond to past research that show growth impacts at DMR ≥ 3 (Thomson et al. 1985; Muir and Hennon 2007). The models fit cumulative probabilities of each DMR class simultaneously, respecting their ordinal nature. Four tree-level predictor variables were considered: distance from the edge (m), seed load (a unitless proxy described below), DBH (cm) and crown class. Continuous predictors were scaled by subtracting the mean and dividing by the standard deviation.

Collinearity was assessed in exploratory data analysis and with attention to inflated standard errors (Hosmer, Lemeshow, and Sturdivant 2013). To build models we used an intentional forward-selection approach, with the Akaike Information Criterion (AIC) as a metric for fit and likelihood ratio tests to assess the significance of terms (Agresti 2010). Successively, the following models were fit:

1. Single-variable
2. Two-variable
3. Two-variable with first-order interactions
4. Three-variable
5. Three-variable with first-order interactions
6. Three-variable with first-order interactions and a random site effect

These combinations explained all variable combinations interpreted to have ecologically plausible rationales, which aligns with our inferential modelling objective. During initial model building the complementary log-log link function was tested; this was based on our expectation that high DMR values would be relatively rare at this time point and a left-skewed, asymmetric, cumulative distribution function might provide a better fit. There was no improvement in AIC, and the logit link was retained. The proportional odds assumption was assessed visually and by testing the significance of additional terms that allowed separate effects on each logit for a single predictor (partial proportional odds models) with likelihood ratio tests (Agresti 2010). In the visual tests, logits of each partial proportional odds model were compared across the range of a focal predictor and parallel logits were interpreted to indicate the assumption was met. Predicted probabilities from the partial proportional odds models were compared to a nested proportional odds model to indicate the magnitude of the effect when the assumption was violated.

2.1.4. Seed load

Seed load is a proxy metric for the amount of HDM seed being deposited on the crown of a hemlock tree in the regenerating component by live infected hemlock trees in the mature component. We developed it to provide a finer scale predictor of HDM infection pressure than raw distance from the source. The proxy only considers live mature source trees and therefore reflects the current infection pressure a tree is under. However, given the relatively young age of the regenerating components (23–45 years) it likely captures some of the historical seed pressure on a regenerating tree. Numerous conceptual

ideas in the proxy build from the simulation model of HDM epidemiology developed by Robinson and Geils (2006). The proxy approximates three processes: (1) seed production, (2) seed dispersal and (3) interception. A prior step is required to generate crown volume estimates of source trees in the mature component.

Crown volume was estimated with crown profile equations developed from stand grown hemlock located in northwest Oregon (Marshall, Johnson, and Hann 2003). The stands used to develop the equations are generally younger and have lower basal area than the mature components but the ranges of tree height and DBH represented cover the range in our data. The minimum inputs for the equations are tree height, height to the base of the live crown and DBH. The height components of these inputs were lacking for most trees in our data and were estimated (described below); recognizing the error introduced by the estimations, crown volume, and the inputs that generate it, were estimated for hypothetical average trees in each crown class at each site, then applied to all trees in each combination.

To estimate height, we assumed constant factors relating heights of suppressed and intermediate crown classes to the mean height of dominant and codominant trees (top height), which was measured or estimated from VRI (Table 2). At one site (cr_3), heights were measured from intermediate and suppressed crown classes. Data from this site was used to calculate the ratios of suppressed and intermediate crown class heights to top height. These ratios were applied as factors to the estimates of top height across all sites to get height estimates for suppressed and intermediate crown classes.

To estimate height to live crown, data from the BC Government's permanent sample plot (**PSP**) network (accessed 24 Jul 2024; Forest Analysis and Inventory Branch 2024a) was used to fit a linear mixed effect model with tree height as the explanatory variable (Table 5 and 6; Figure 3). PSP data was filtered by similarity to the mature components based on proximity (using Timber Supply Area boundaries), biogeoclimatic subzones/variants and hemlock top height (within one standard deviation of the mean across mature components). PSP data contains multiple measurements of established sites over time; no remeasurements were present in the data used to build models so each site represented a single time point. A simple linear model was attempted first but did not meet the equal variance assumption. The

model with a random site effect met all assumptions. We applied this model to our data, assuming they were the population average (random effect = 0) to get height to live crown estimates.

Table 5. Summary of BC Ministry of Forests permanent sample plot dataset (PSP) used to fit height to live crown model. PSP sites in our study area and in the right climate zones were matched to the mature component in our dataset (HDM) based on similarity in hemlock top height; top height estimates from HDM dataset are summarized in study site descriptions but included here for comparison. Model predicts height to live crown from tree height. N_{site} = number of sites in dataset, N_t = number of individual trees.

Dataset	N_{site}	N_t	Top height (m)			Height all live hemlock (m)			Height to live crown(m)		
			Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
HDM	11	80	34.8	23.7	44.2	-	-	-	-	-	-
PSP	14	308	35.7	29.8	39.6	22.4	2.0	47.8	14.2	1.0	34.0

Table 6. Summary of linear mixed effect model predicting height to live crown (m) from tree height (m) for hemlock trees in the mature component. Error partitioning between tree and site (random effect) is described by standard deviation of the errors (SD(error)). SE = standard error, DF = degrees of freedom.

Term	Site	Tree			
SD(error)	2.86	2.49	-	-	-
Coefficients	Estimates	SE	DF	t-value	p-value
Intercept	0.680	0.850	293	0.800	0.424
Tree height	0.559	0.0134	293	41.8	<0.0001

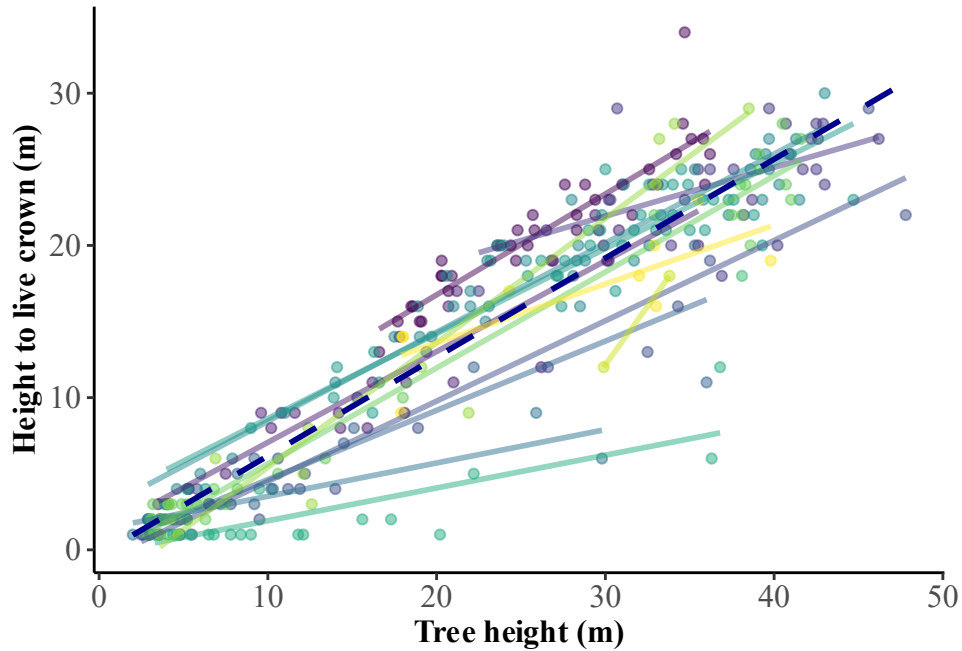


Figure 3. Relationship between height to live crown and tree height in hemlock trees in the BC Government Permanent Sample Plot data. Colours are individual sites. Blue, dashed line is best fit line, corresponding to population average (random effect = 0), which is used to generate predictions for trees in the mature components in our data.

To generate the final input, DBH was averaged within each site-crown class combination. The inputs (DBH, tree height, height to live crown) were fed into equations for maximum crown width of an open grown tree, largest crown width of stand grown tree, distance above live crown where the largest crown width occurs and crown width at a given height from Marshall, Johnson and Hann (2003). This generated tree profiles which were integrated over crown length to obtain crown volume (Table 7).

Table 7. Summary of crown volume and underlying tree dimensions, for hemlock trees in the mature component. Tree dimensions are summarized from field measurements or VRI (DBH, height of D/C crown class) or estimated (height of I and S crown classes, height to live crown, maximum crown width, crown volume) for each site-crown class combination and then applied to all trees in that combination. F_{HT} were estimated from a single site (cr_3) where all crown classes were measured. CC = crown class, D/C = combined dominant and codominant, I = intermediate, S = suppressed, F_{HT} = factor relating height of each crown class to height of D/C crown class.

CC	F_{HT}	DBH (cm)			Height (m)			Height to live crown (m)			Max crown width (m)			Crown volume (m^3)		
		Median	Min	Max	Median	Min	Max	Median	Min	Max	Median	Min	Max	Median	Min	Max
D/C	-	51.7	37.6	90.9	35	24	44	20	14	25	7.7	6.1	11	310	150	840
I	0.60	23.8	13.8	33.9	21	14	27	13	8.7	16	4.6	3.1	5.9	63	39	130
S	0.33	14.4	10.9	19.4	12	7.8	15	7.1	5.0	8.8	3.4	2.9	4.1	19	10	28

Seed production (SP) for a given source tree is estimated similarly to Robinson and Geils (2006) as,

$$SP = \sum_{t=1}^3 (CV_t DMR_t) \quad (1)$$

where t identifies the crown third in the set of thirds $I = \{1, 2, 3\}$, CV_t is crown volume, and DMR_t is dwarf mistletoe rating. Crown volume is multiplied by DMR (0, 1 or 2) for each crown third and then summed for the tree. Seed dispersal is modelled with a modified exponential decay function as,

$$PSP = \begin{cases} 0.378, & \text{if } BD \leq LCW \\ 0.378 e^{-0.25ED}, & \text{if } BD > LCW \end{cases} \quad (2)$$

where PSP is the proportion of seed reaching a given distance from the source tree bole (BD), LCW is the crown width of the source tree at its widest point and ED is the distance from the edge of the source tree crown. Locations within the crown width of the source tree, are assumed to be exposed to a constant proportion of the seed production value. Beyond the crown edge, the proportion declines to zero (Figure 4A). This function is matched to data from a study that trapped HDM seed at variable distances from an infected source tree (Smith 1966).

Interception of HDM seed is estimated by identifying intervening trees on the path between a source tree in the mature component and a target tree in the regenerating component. Only other trees in the mature component were considered significant sources of interception, so intervening regenerating trees are ignored. Lines connecting each pair are generated (Figure 4C) and then buffered to 5 m width. Intervening trees in the mature component that intersect the buffered area are counted by crown class (Figure 4D). Each intervening tree reduces seed load by a factor; these factors were defined for each source tree-intervening tree-crown class combination, using results from interception experiments in Bloomberg et al. (1980) as a guideline (Table 8).

Table 8. Interception factors for source tree-intervening tree pairs. Factors are applied to reduce a given seed load value. For example, for a D/C-I pair with a seed load value ignoring interception of 100, seed load with interception is $= 100 - (100 * 0.45) = 55$. Abbreviations: D/C = dominant/codominant, I = intermediate, S = suppressed.

Source tree crown class	Intervening tree crown class	Factor
D/C	D/C	0.90
D/C	I	0.45
D/C	S	0.00
I	D/C	0.90
I	I	0.90
I	S	0.45
S	D/C	0.00
S	I	0.90
S	S	0.90

Seed production, dispersal and interception combine to estimate seed load. For a given source tree in the mature component and target tree in the regenerating component, the workflow is as follows. Seed production is estimated for the source tree with equation (1). A path between the trees is generated, which gives their separation distance. This functions as the input to the dispersal equation (2), which estimates the seed load in the absence of interception. Intervening trees are identified, and the seed load value is factored by their cumulative interception to give a final estimate. This workflow occurs for all pairs of source and target trees within 25.0 m of each other. Total seed load for a given regenerating tree, is the sum of the seed load from all source trees it is affected by (Figure 4E and 4F). Seed production and seed load are summarized in Table 9.

Table 9. Summary of seed production and seed load proxies. Seed production is summarized across all live hemlock trees in the mature component. Seed load is summarized for all live hemlock trees in the regenerating component within 25.0 m of ≥ 1 source tree in the mature component (live hemlock with $DMR \geq 1$); this reflects the distribution of seed load in trees within the dispersal distance of source trees and excludes trees > 25.0 m from all source trees that have a seed load = 0, by definition.

Variable	Dominant/codominant			Intermediate			Suppressed		
	Median	Min	Max	Median	Min	Max	Median	Min	Max
Seed production, mature hemlock	380	0.00	1700	18	0.00	180	10	0.00	47
Seed load, regen hemlock, no interception	27	0.022	1400	27	0.023	1000	24	0.034	1300
Seed load, regen hemlock, with interception	14	0.00	470	14	0.00	230	10	0.00	540
Proportion seed load intercepted	0.57	0.00	1.0	0.45	0.00	1.0	0.52	0.00	1.0

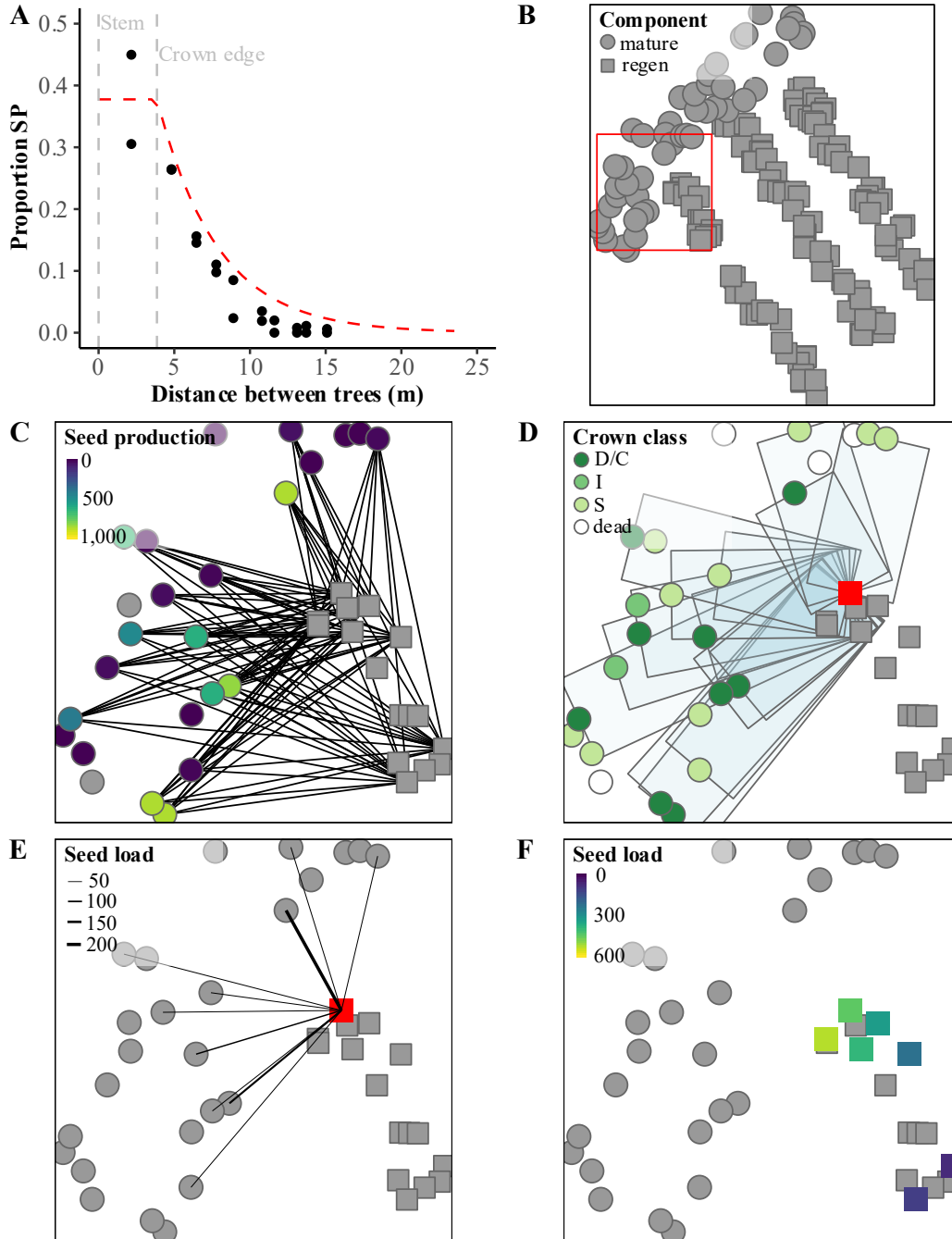


Figure 4. Summary of steps to calculate seed load proxy variable. (A) Dispersal function defines the proportion of the seed production (SP) value (calculated from crown volume and dwarf mistletoe rating) of a live, mature hemlock tree that reaches a given distance. Points are from field study (Smith 1966) that function is based on. (B) Inset area at one research site used in C-F. (C) All pairs of live, infected hemlock trees in the mature component and live hemlock trees in the regenerating component within 25.0 m of each other are located. (D) 5.0 m wide paths are created for each pair and interception is calculated from counts of trees in the mature component by crown class on each path. (E) Seed load is calculated for each pair by combining seed production, a dispersal function (A) and interception (D). (F) Sum of seed load from all pairs gives value for each regenerating tree. Shapes distinguish component in panels B-F. Panels D and E show data for a single focal tree (red). D/C = dominant and codominant, I = intermediate, S = suppressed.

2.2. Technoeconomic analysis of hemlock's role in timber supply

An existing forest estate model developed for two managed forest areas near the community of Pemberton BC was used for the analysis—the Spel kúmtn Community Forest (SCF), which is co-managed by the Lil'wat First Nation and the Village of Pemberton, and the adjacent First Nation Woodland Licence (FNWL) (Man 2025). SCF covers ~17 000 ha, of which ~10 000 ha is forested and ~3 000 ha is allocated to timber harvesting. The FNWL is larger, covers ~142 000 ha, contains ~63 000 ha of forested land and ~16 000 ha of forested land allocated to harvesting. The areas allocated to timber harvesting are identified from the total forest area by removing subsets for various land use considerations (ownership, cultural and ecological protected areas, wildlife habitat) and filtering the remaining area to productive accessible stands. Of the total forested area, 61% of SCF and 56% of the FNWL are in the CWH, which is a coarse proxy for sites where hemlock is present and HDM is hazard is high (Rusch et al. 2019).

The forest estate model is built in the Patchworks software (Spatial Planning Systems, n.d.). Forest conditions are projected over 300 years, in 10-year time steps, iteratively, with the model seeking to optimize user-defined targets and goals. The Variable Density Yield Prediction and Table Interpolation for Stand Yields models (Penner 2021) are used to predict natural and managed stand growth, respectively. We applied a base case scenario that aimed to maximize yield, with the limitation that harvest did not deplete the available volume below the initial conditions over the last five time steps. The model assumes a clearcut with reserves silvicultural system is applied for all harvests. Regeneration is as a combination of planting and natural infill; for hemlock, natural infill contributes proportionally more. To predict value creation potential (VCP; a proxy for revenue) from projected volume, log prices were retrieved for the BC coast region for the three-month period ending December 2024 (BC Ministry of Forests 2024a). A price for each species was assigned by averaging the price of the highest grade and the utility grade (Table 10). Species without a reported price were assigned an average, calculated by dividing the total market value by the total trading volume. Harvesting cost was set at \$30/m³; this covers direct

harvesting and omits secondary costs (e.g. new road construction, secondary transportation). Value creation potential was calculated as the difference between projected income and harvesting costs.

Table 10. Log prices by species used in technoeconomic analysis. Model price is the average of the prices for the highest grade and the utility grade (U). Species not listed are assigned the market average.

Species	Highest grade price (\$/m ³)	U grade price (\$/m ³)	Model price (\$/m ³)
Cedar	595.74	125.57	360.66
Douglas-fir	408.62	80.92	244.77
Hemlock	120.97	55.60	88.29
Amabilis fir	120.97	55.60	88.29
Spruce	84.10	54.81	69.46
Other	-	-	126.20

Chapter 3: Results

3.1. Site-level summaries of infection

HDM infection varied substantially over the range of age and conditions represented in the edge infection study sites (Figure 5 and 6A). In the mature component, incidence of infection was consistently high, but severity and the amount of infected host material (represented by infected basal area) ranged widely (Figure 6; Table 11). The mean and first and third quartiles (**Q1** and **Q3**) of incidence in live hemlock trees in the dominant/codominant crown class were 95%, 95% and 100%. For the same tree group, the means and quartiles of average DMR were 4.1, 3.3 and 5.2; for DMI they were 4.3, 3.3 and 5.4. Total infected basal area of live and dead hemlock trees ranged from 18 to 106 m²/ha (mean = 53 m²/ha); this variation was driven by differences in the abundance and size distribution of hemlock trees (Figure 7). Sites with severe infection and large median hemlock tree sizes (e.g. mi_1, cr_3), and contrasting sites with smaller median hemlock tree sizes and light to moderate infection (e.g. mk_1, ph_1), were both represented. Other sites fell between these extremes. Witches' brooms were common in live hemlock trees, particularly in dominant/codominant crown class (mean = 82%). Incidence of stem infections averaged 31% for the live hemlock in the dominant/codominant crown class.

In the regenerating component, most infected trees occurred within 15.0 m of the edge (Figure 8); the infection perimeter metric (distance from edge that captures 90% of infected trees), averaged 14.1 m (median = 12.4 m, range = 2.7–32.0 m). In live dominant/codominant hemlock trees within 15.0 m of the edge, infection incidence was relatively high (mean = 69%, Q1 = 61%, Q3 = 91%) but severities were light (average DMR: mean = 1.2, Q1 = 0.5, Q3 = 1.6; DMI: mean = 1.9, Q1 = 1.6, Q3 = 2.0) (Figure 6; Table 12). A modest proportion of trees in this group (which represent future hemlock crop trees) have DMR ≥ 3, where growth impacts occur (mean = 17%, Q1 = 0 %, Q3 = 20%); these trees were located a maximum of 13.5 m from the edge and occurred at six sites. All infection metrics except for DMI, which only considers infected trees, were reduced in the 0.0–50.0 m interval, relative to the 0.0–15.0 m interval;

this difference was largely a dilution effect of adding healthy trees, because of the concentration of infected trees near the edge. For the live dominant/codominant hemlock tree group, incidence averaged 31% (Q1 = 21%, Q3 = 39%) and severity metrics (average DMR and DMI) averaged 0.4 (Q1 = 0.2, Q3 = 0.6) and 1.7 (Q1 = 1.4, Q3 = 2.0), respectively. Trends associated with crown class were apparent. Dominant/codominant and intermediate classes had higher rates of infection and severity than trees in the suppressed class over both intervals. The incidence of witches' brooms and stem infections followed patterns of infection severity, averaging 21% and 14% in live, dominant/codominant within 15.0 m of the edge. Vertical distribution of infection in the regenerating forest component followed a bottom-up pattern, with infection concentrated in lower and middle crown thirds (Figure 9). Outlying sites of both high (e.g. cr_3) and low infection (ph_1) were apparent. At the cr_3 site, the infection perimeter was 32.0 m, 100% live dominant/codominant hemlock were infected within 15.0 m of the edge and average DMR and DMI (equivalent in this case) were 3.0. At the ph_1 site, only a single infected tree was located in the regenerating component.

At the site where amabilis fir was present (ph_2), its abundance was similar to hemlock. In the mature component, it made up 45 % of all trees and 9 % of basal area (compared to hemlock, which made up 40 % of trees and 10 % of basal area). Large (92.0–280.0 cm DBH) cedar were present in the mature component of this site and accounted for most of the site basal area. In the regenerating component, 23 % of trees and 26 % of basal area was amabilis fir; comparatively, hemlock made up 30 % and 33 %. HDM incidence in live trees, average DMR and DMI for amabilis fir in the mature component were 47 %, 0.7 and 1.5; for the same metrics, hemlock was 85 %, 2.0 and 2.4. In the regenerating component, incidence in live trees, average DMR and DMI were 6 %, 0.1 and 1.0 for *A. amabilis* and 27 %, 0.3 and 1.2 for hemlock.

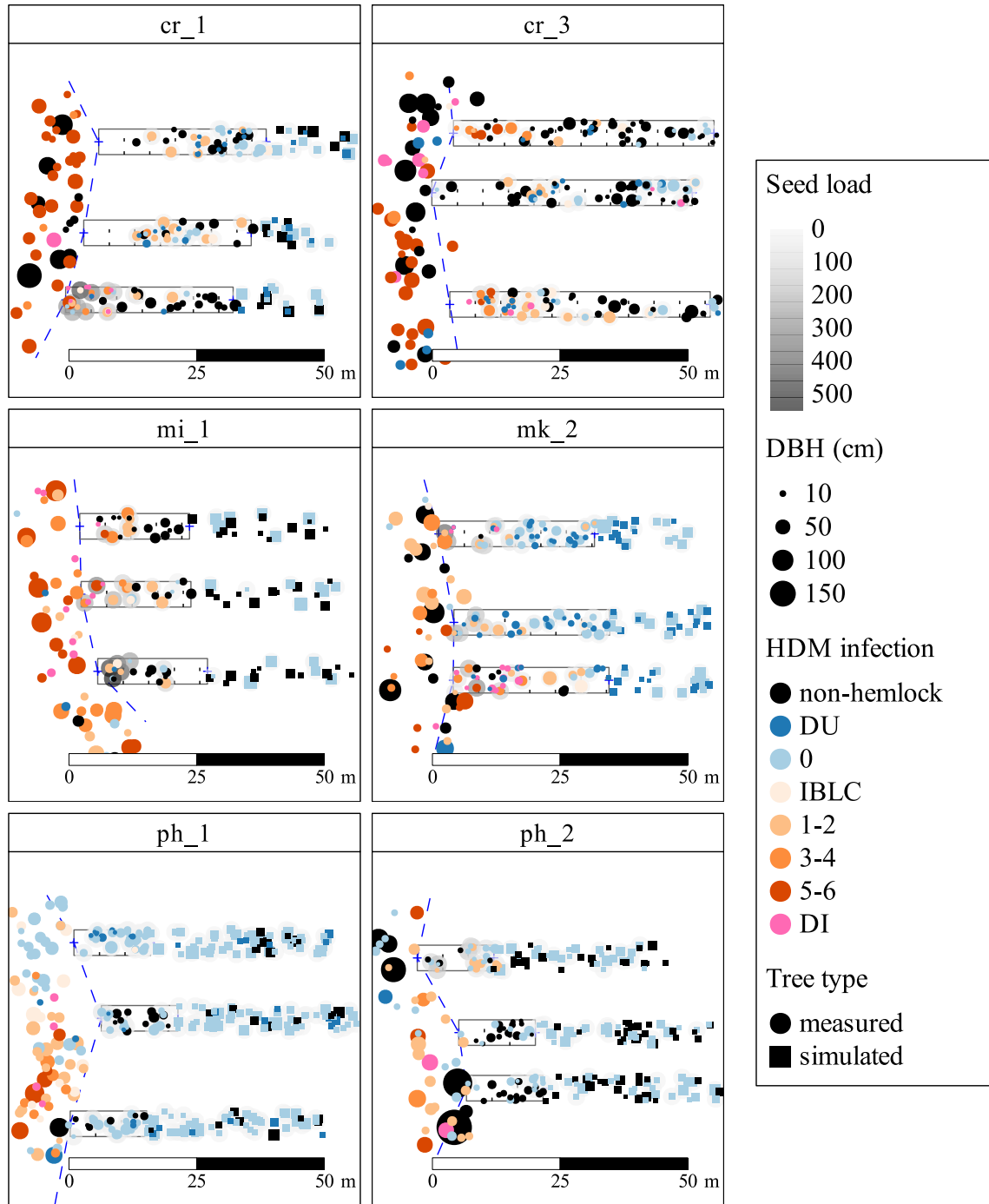


Figure 5. Stem maps showing hemlock dwarf mistletoe (HDM) infection at six representative sites. Mature forest component is on left side in each panel. Regenerating component is on the right. Live hemlock trees in regenerating component are overlaid on a background of seed load – a proxy for the amount of HDM seed deposited on each tree from infected hemlock in the mature component. For HDM infection, dead trees are recorded on a presence/absence scale (DU = dead uninfected, DI = dead infected) and live trees are recorded on the dwarf mistletoe rating (DMR) scale (0, IBLC = infected below live crown, 1–6).

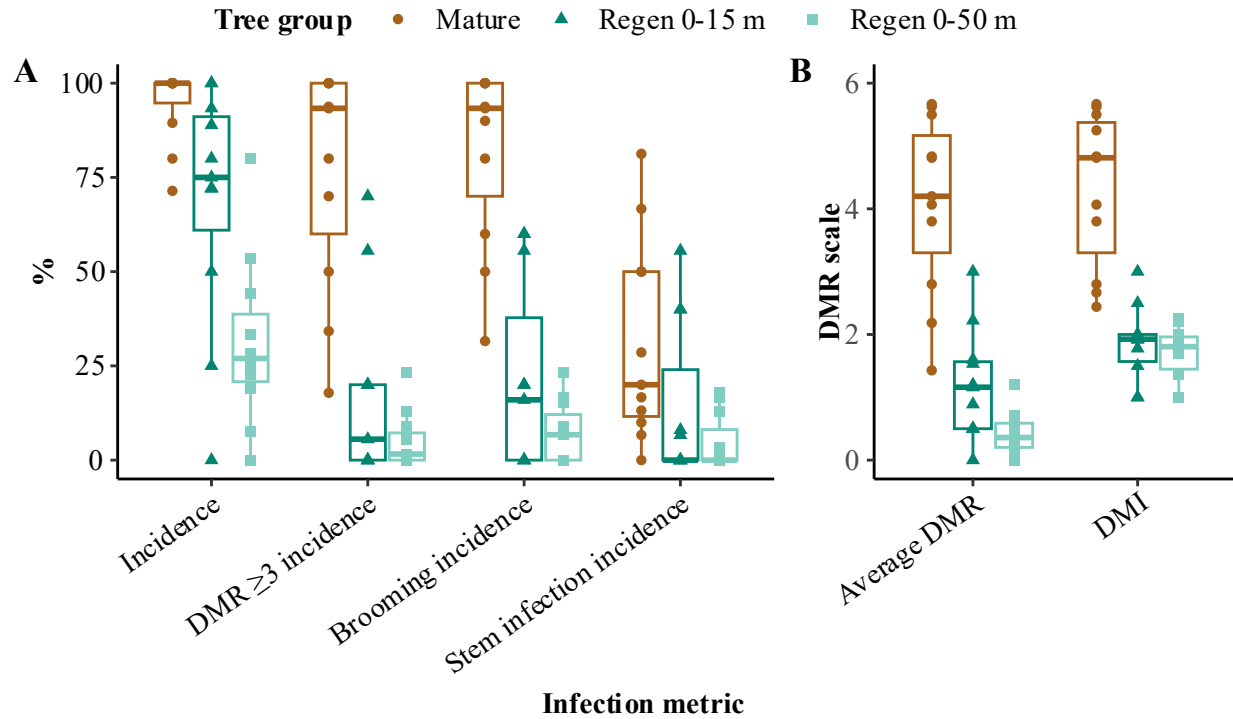


Table 11. Summary metrics of hemlock dwarf mistletoe infection in live hemlock trees in the mature component (infection source). Data are calculated at the site level (e.g. % trees infected is calculated for each site and range shows spread of these values). DMR ≥ 3 corresponds to infection severity where trees experience impacts to height and diameter growth. Average DMR is the average rating across all live hemlock trees. DMI (dwarf mistletoe index) is the average rating across all live hemlock trees with DMR ≥ 1 . DMR = dwarf mistletoe rating, CC = crown class, D/C = dominant and codominant, I = intermediate, S = suppressed.

CC	% trees infected			% DMR ≥ 3			% brooms			% stem infection			Average DMR			DMI		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
all	85	36	100	55	7	100	55	8	100	31	2	83	3.0	0.6	5.4	3.5	2.0	5.4
D/C	95	71	100	76	18	100	82	32	100	31	0	81	4.1	1.4	5.7	4.3	2.4	5.7
I	87	37	100	54	0	100	57	2	100	36	0	100	2.9	0.4	5.2	3.2	1.0	5.2
S	75	9	100	31	0	100	30	0	100	22	0	100	1.9	0.2	5.0	2.6	1.0	5.0

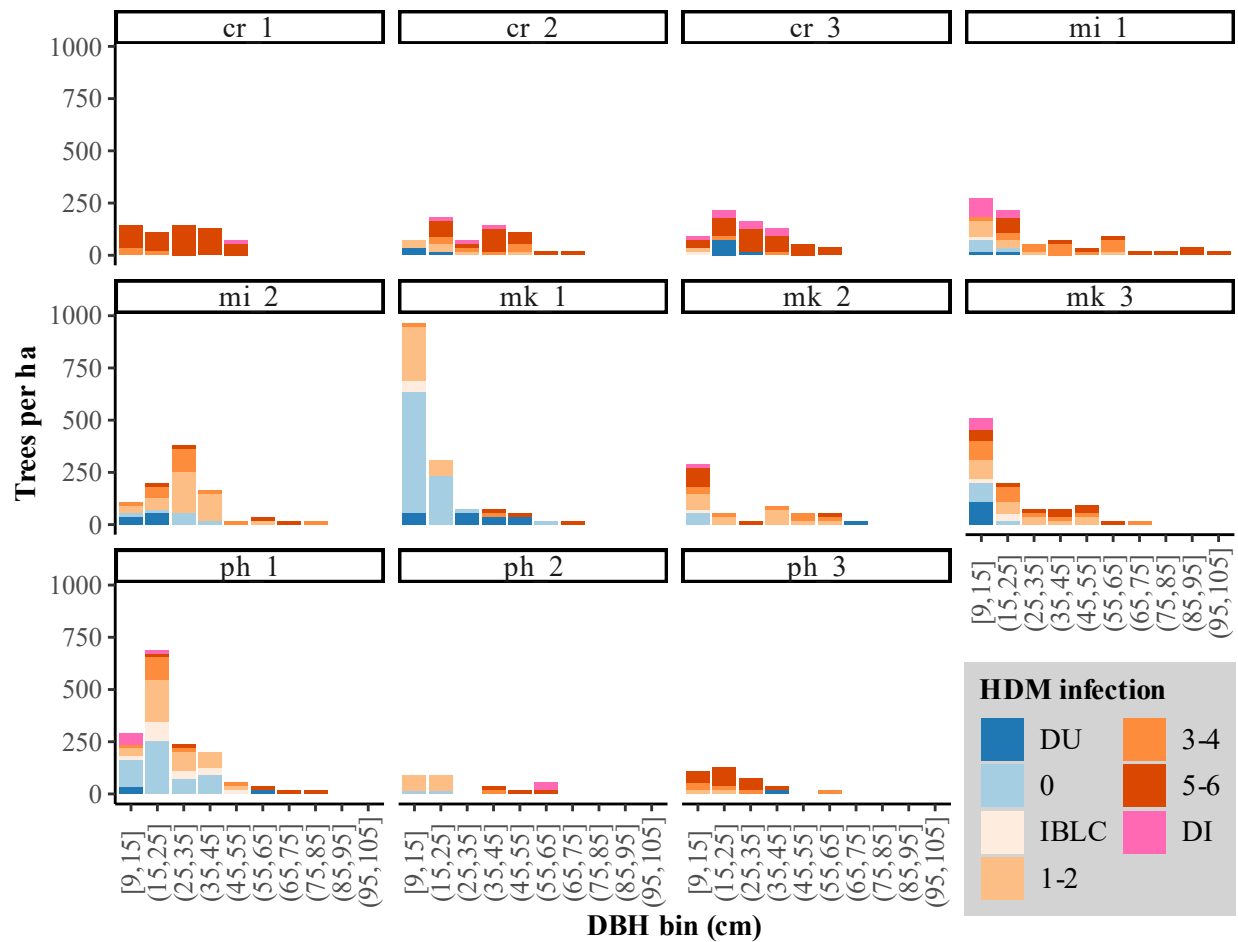


Figure 7. Distribution of hemlock dwarf mistletoe (HDM) infection in hemlock trees by DBH in the mature forest components at each site. For HDM infection, dead trees are recorded on a presence/absence scale (DU = dead uninfected, DI = dead infected) and live trees are recorded on the dwarf mistletoe rating (DMR) scale (0, IBLC = infected below live crown, 1–6).

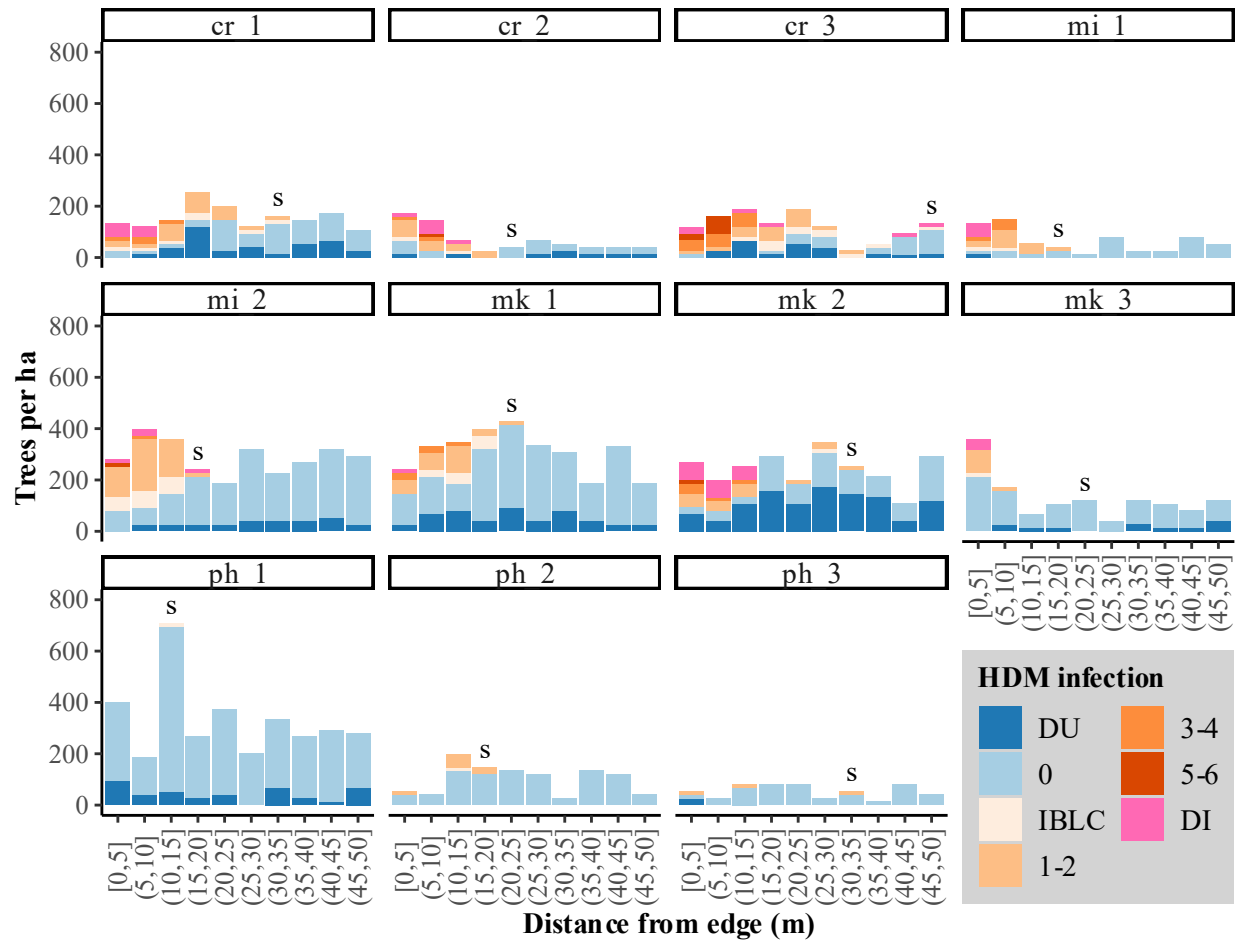


Figure 8. Hemlock dwarf mistletoe (HDM) infection in hemlock trees in the regenerating forest component in 5.0 m intervals from the edge. For HDM infection, dead trees are recorded on a presence/absence scale (DU = dead uninfected, DI = dead infected) and live trees are recorded on the dwarf mistletoe rating (DMR) scale (0, IBLC = infected below live crown, 1–6). “s” indicates the last interval with measured trees on at least one transect at a site; intervals beyond this contain only simulated data.

Table 12. Summary metrics of hemlock dwarf mistletoe infection in live hemlock trees in the regenerating component. Data are calculated at the site level (e.g. % trees infected is calculated for each site and range shows spread of these values). Summary is organized into two intervals (Int) based on distance from the mature forest component (infection source) and by crown class (CC). 0.0–15.0 m interval only contains measured trees and is under the influence of overstory HDM seed from the mature component. 0.0–50.0 m contains measured and simulated trees and includes regions beyond theoretical seed dispersal from mature component. DMR ≥ 3 corresponds to infection severity where trees experience impacts to height and diameter growth. Average DMR is the average rating across all live hemlock trees. DMI (dwarf mistletoe index) is the average rating across all live hemlock trees with DMR ≥ 1 . DMR = dwarf mistletoe rating, D/C = dominant and codominant, I = intermediate, S = suppressed.

Int (m)	CC	% trees infected			% DMR ≥ 3			% brooms			% stem infection			Average DMR			DMI		
		Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
0.0–15.0	all	54	1	95	15	0	74	13	0	58	11	0	32	0.9	0.0	2.9	1.8	1.0	3.2
0.0–15.0	C	69	0	100	17	0	70	21	0	60	14	0	56	1.2	0.0	3.0	1.9	1.0	3.0
0.0–15.0	I	50	0	100	20	0	80	11	0	50	8	0	50	1.0	0.0	2.8	2.1	1.3	3.5
0.0–15.0	S	39	0	75	9	0	75	8	0	75	9	0	50	0.6	0.0	2.8	1.7	1.0	3.7
0.0–50.0	all	27	0	70	5	0	26	5	0	23	4	0	15	0.4	0.0	1.2	1.7	1.0	2.5
0.0–50.0	C	31	0	80	5	0	23	7	0	23	5	0	18	0.4	0.0	1.2	1.7	1.0	2.3
0.0–50.0	I	30	0	100	11	0	50	9	0	50	6	0	50	0.6	0.0	2.3	1.9	1.0	3.5
0.0–50.0	S	20	0	55	4	0	27	3	0	27	4	0	18	0.3	0.0	1.3	1.5	1.0	2.3

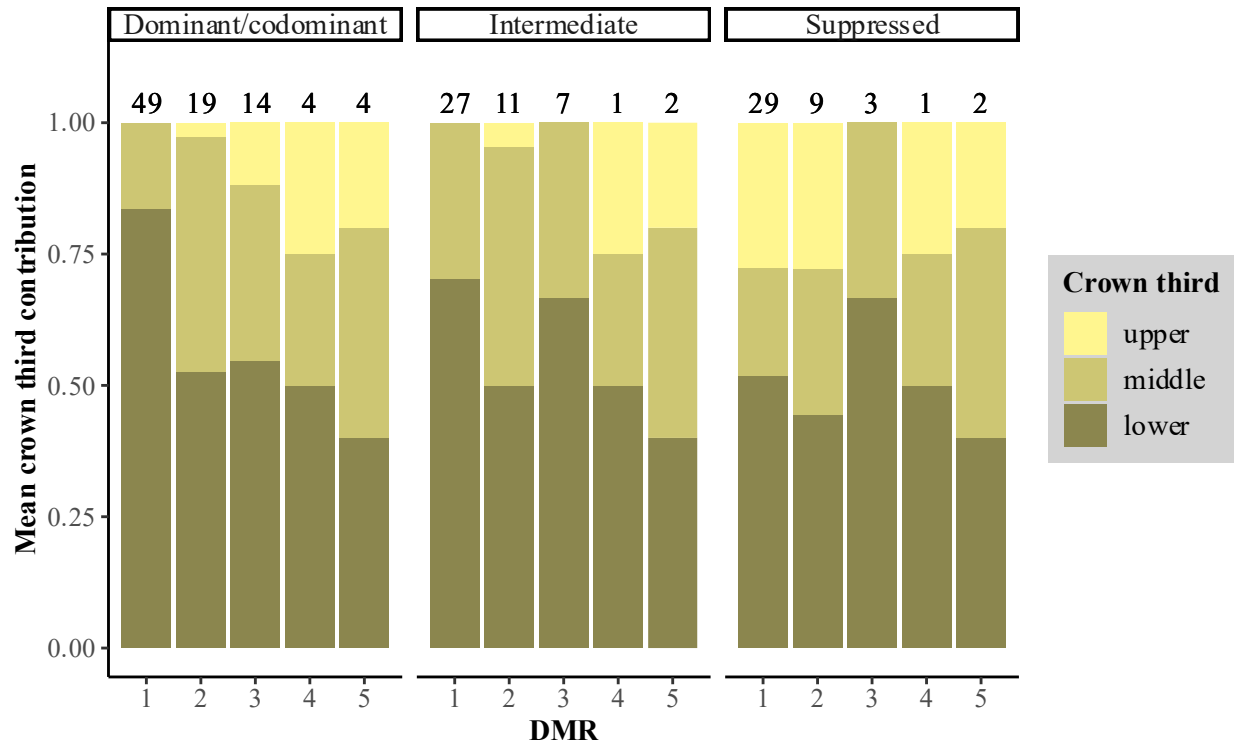


Figure 9. Mean contribution of each crown third to tree-level dwarf mistletoe rating (DMR) in regenerating forest component. Data are from live hemlock trees with DMR ≥ 1 . Panels separate trees by crown class. Crown third contribution is the proportion of tree DMR (bounds: 1–6) that comes from each crown third (bounds: 0–2) (e.g. for a tree with DMR = 3, the maximum contribution of a given crown third = $2/3 = 0.66$); patterns reflect the vertical distribution of infection. Labels at top of bars are the sample size.

3.2. Modelling results

The dataset used for modelling consisted of 1093 live hemlock trees, 50 % of which were simulated, healthy ($DMR = 0$), trees beyond the outer infection limit of a site (Table 13). All candidate tree-level variables showed some relationship with DMR in exploratory data analysis (Figure A7). The form of the selected model relating DMR to tree-level variables was,

$$\ln \frac{P(DMR_{m,i} \leq j)}{1 - P(DMR_{m,i} \leq j)} = \theta_j - \beta_1 DIST_{s,i} - \beta_2 SL_{s,i} - \beta_3 DBH_{s,i} - \beta_4 (DIST_{s,i} SL_{s,i}) - \beta_5 (DBH_{s,i} SL_{s,i}) + u_m \quad (3)$$

$DMR_{m,i}$ is DMR collapsed to three levels (j), θ_j are intercept coefficients, $DIST_{s,i}$ is distance to the edge (m), $SL_{s,i}$ is seed load (unitless), u_j is a random site effect, s indicates a variable has been scaled and i and m indicate the observation and the site, respectively. Model coefficients are reported in Table 14.

Guidance from the documentation of the package used to fit the models (ordinal; Christensen 2019) was used to interpret the Hessian condition number, which measures model identifiability. Numbers $>10^4$ indicate problems; the value for the final model (430) indicated it was well defined.

Table 13. Summary of the modelling dataset. Includes of all live measured and simulated hemlock trees in the regenerating component. Organized by to show distribution of sample size and predictor variables across dwarf mistletoe rating (DMR) classes. Note, final model uses collapsed version of adapted DMR: 0, IBLC–2, 3–5. IBLC = infected below live crown, N = number of trees, N_m = number of measured trees, N_s = number of simulated trees.

Group	N	N_m	N_s	Distance from edge (m)			Seed load (no units)			DBH (cm)		
				Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
All	1093	543	550	23.4	0.1	50.0	24	0	543	12.8	4.0	41.0
0	862	312	550	26.8	0.1	50.0	10	0	543	12.3	4.0	41.0
IBLC	49	49	0	14.7	0.3	48.9	58	0	442	13.9	4.5	32.7
1	105	105	0	11.6	0.1	32.7	76	0	468	13.9	4.0	35.4
2	39	39	0	8.8	0.2	20.9	66	1	395	15.7	4.3	36.7
3	24	24	0	7.2	0.6	14.4	96	0	309	18.3	6.7	33.0
4	6	6	0	5.0	1.9	8.2	240	69	441	19.7	11.3	29.5
5	8	8	0	6.3	3.0	8.6	80	0	288	17.5	10.7	25.5

Table 14. Summary of final model relating tree-level predictors to dwarf mistletoe rating (DMR). Variables: DMR = dwarf mistletoe rating collapsed to three levels (0, IBLC–2, 3–5; denoted with subscript j), DIST = horizontal distance from the mature-regenerating forest interface (m), SL = seed load (unitless), DBH = diameter at breast height (cm). Interactions terms are identified as: variable1*variable2. The subscript s indicates variables were scaled. Tests of significance for coefficients are Wald’s tests; more appropriate likelihood ratio tests are included in the Appendix. IBLC = infected below live crown, SD = standard deviation, SE = standard error.

Convergence and Random Effects				
Hessian condition	430.	-	-	-
Random effect SD	1.98	-	-	-
Term	Estimate	SE	z-statistic	p-value
Intercept coefficients (θ_j)				
0 IBLC–2	2.04	0.654	-	-
IBLC–2 3–5	6.21	0.732	-	-
Model coefficients (β_{1-5})				
DIST _s	-1.96	0.279	-7.02	<0.0001
SL _s	0.889	0.126	7.04	<0.0001
DBH _s	2.87	0.543	5.28	<0.0001
DIST _s *SL _s	1.73	0.378	4.58	<0.0001
DBH _s *SL _s	0.476	0.118	4.02	0.0001

Crown class, the only tree-level variable considered that is not in the selected model, was dropped because it is colinear with DBH (Figure A8). The variables gave similar AIC values in univariate models, and DBH was selected for ease of working with a continuous variable (Table A5, compare m6 and m7). Seed load has some structural collinearity with distance from the edge but contains distinct ecological information; no inflated standard errors were observed when both variables were present, so models with both variables were interpreted as valid. Model fit statistics are reported in Table A5 and likelihood ratio tests of term significance are in Table A6. The proportional odds assumption was met for all terms except distance from the edge (Table A6), but with visual inspection we interpreted this deviation to tolerable (Figure A9). The predicted probabilities of the partial proportional odds model for distance from the edge diverged from the proportional odds model near the edge (0.0–7.5 m) (Figure A10); disagreement between the predicted probabilities was <0.1 for most observations and did not result in a change in the order of the most probable DMR class. The range of log-likelihood and AIC values between the null model and the final model were [-677, -331] and [1338, 679], respectively. The largest jumps in log-likelihood and AIC were from the addition of new variables and the random site effect; interaction terms had smaller effects.

The probability of infected DMR classes (IBLC-2, 3–5) decrease with increasing distance from the edge and decreasing DBH, but these variable effects depend on seed load (Figure 10 and A11). Large trees that are close to the edge (e.g. <15.0 m) and under high seed load have the highest probability of infected DMR classes. Small trees distant from the edge (and by function of being far from the edge, under no seed load) have the highest probability being uninfected (DMR = 0). Random site effects are substantial and can reverse the most probable DMR class for an average tree when large (Figure 11 and A12).

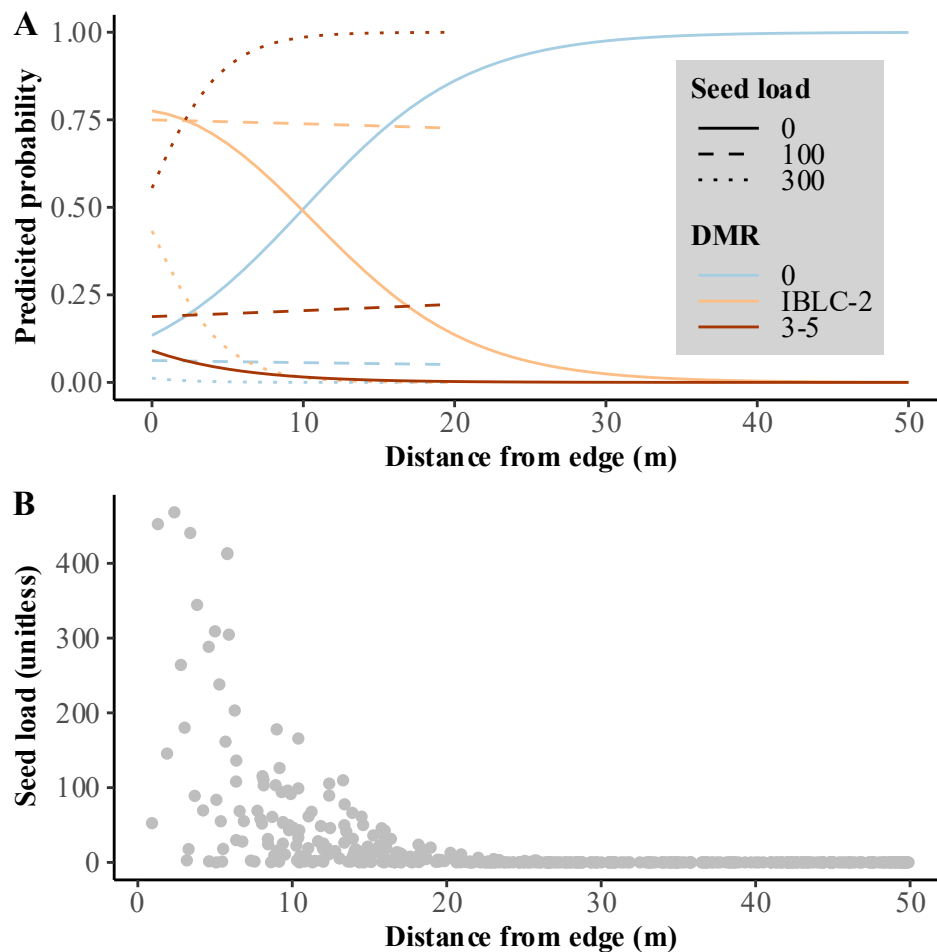


Figure 10. (A) Effect of increasing distance from the edge on predicted probabilities of dwarf mistletoe rating (DMR) classes for live dominant/codominant hemlock. (B) Seed load values observed in raw data with increasing distance from the edge. Within 20.0 m of the edge, the relative probabilities of each DMR class depend on both distance and seed load. Beyond 20.0 m, seed load is ~ 0 and probabilities depend only on distance. High and moderate seed load levels are not plotted in beyond 20.0 m in (A) for this reason. DBH is set at the mean value of dominant/codominant trees. IBLC = infected below live crown.

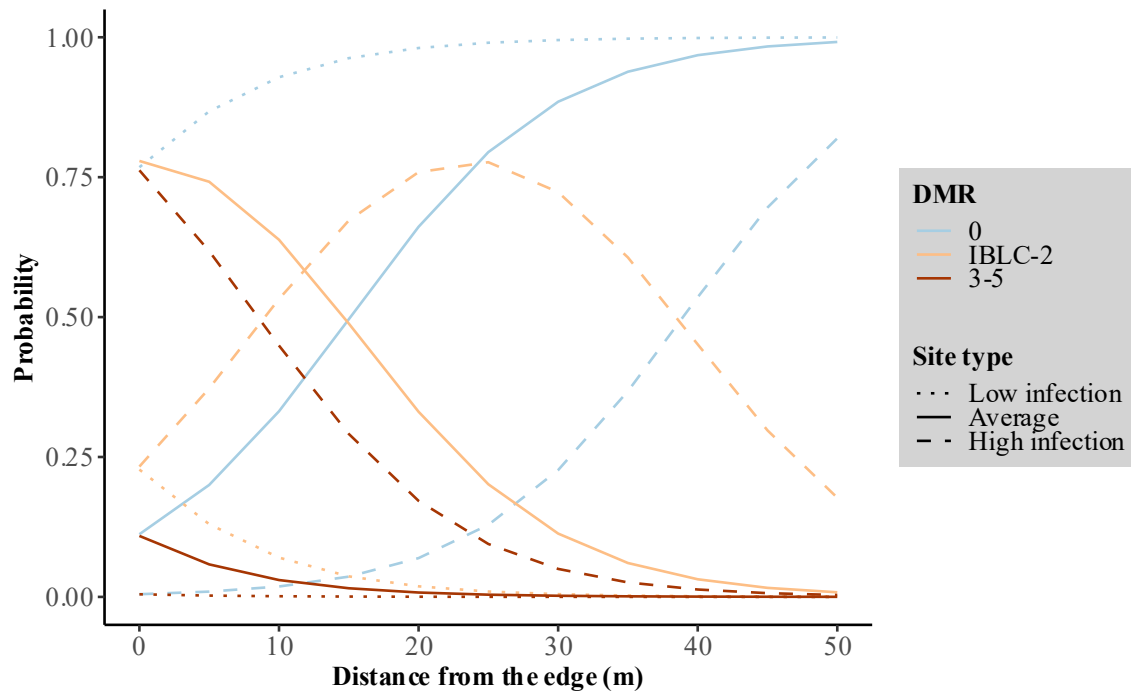


Figure 11. Between site variation in predicted probabilities of dwarf mistletoe rating (DMR) classes for dominant/codominant hemlock. Probabilities are shown for hypothetical sites that have average infection (random effect = 0), high infection (large positive random effect) and low infection (large negative random effect) patterns. Seed load is set at its mean value. Note: this is unrepresentative beyond 20.0 m because seed load is ~ 0 . These distances have been retained to illustrate the magnitude of the random site effect. IBLC = infected below live crown.

3.3. Hemlock's contribution to timber supply

Results summarizing the projected timber supply by species in the two case study areas are in Table 15. The SCF area sustained a harvest volume of 14 000 m³ per year from its 3 000 ha timber harvesting land base. The larger FNWL sustained a rate of 78 000 m³ per year from its 16 000 ha timber harvesting land base. Trends across the 300 years simulated were stable and thus results are reported for the entire period. Hemlock made up a moderate amount of the volume harvested ($\sim 15\%$) and a proportionally smaller amount of the VCP ($\sim 5\%$). The difference between hemlock's share of harvested volume and VCP is driven by its relatively low price ($\$88/\text{m}^3$). This contrasts high-value Douglas-fir ($\$245/\text{m}^3$) and cedar ($\$361/\text{m}^3$), which show the opposite relationship between contributions to volume and VCP. Cedar made up $\sim 18\%$ of volume across the two areas and $\sim 32\%$ of VCP. Douglas-fir contributed $\sim 51\%$ of volume and 58% of VCP.

Table 15. Summary of timber supply and value creation potential (VCP) by major species in the two managed forest areas used in the case study. Simulation projected for 300 years in 10-year periods. Total volume and revenue statistics are for the entire 300 years. All species row combines species listed in table and other minor species not listed explicitly. SCF = Spel kúmtn Community Forest, FNWL = First Nation Woodland Licence.

Area	Species	Total volume (m^3)	Mean volume/10 years (m^3)	Total revenue (\$)	% total volume	% total VCP
SCF	amabilis fir	3.1×10^5	1.0×10^4	1.8×10^7	7%	2%
	cedar	7.7×10^5	2.6×10^4	2.5×10^8	18%	30%
	Douglas-fir	2.4×10^6	8.0×10^4	5.2×10^8	56%	62%
	hemlock	6.3×10^5	2.1×10^4	3.7×10^7	15%	4%
	all	4.3×10^6	1.4×10^4	8.4×10^8	100%	100%
FNWL	amabilis fir	3.7×10^6	1.2×10^5	2.2×10^8	16%	5%
	cedar	4.2×10^6	1.4×10^5	1.4×10^9	18%	33%
	Douglas-fir	1.0×10^7	3.4×10^5	2.2×10^9	44%	54%
	hemlock	3.6×10^6	1.2×10^5	2.1×10^8	15%	5%
	all	2.3×10^7	7.8×10^5	4.1×10^9	100%	100%

Chapter 4: Discussion

The decision to use group retention silvicultural systems in stands with HDM involves weighing the timber impacts of HDM infection against two other factors: (1) the non-timber impacts of HDM infection and (2) the impacts of retention patches. The additional timber impact (i.e. relative to an even-aged clearcut) of HDM in group retention comes from spread into the regenerating stand from infected overstory trees in retention patches and from continued impacts to infected trees in retention patches. The retained trees in variable retention can be selected for their non-timber qualities (e.g. nesting habitat) (Beese et al. 2019), so management guidelines have focused primarily on the additional impacts to regeneration. The results from our field study provide a first indication of infection levels and severities that can be expected in regeneration adjacent to infected edges. Moving from this initial data to a full characterization of the HDM timber impact to regeneration in group retention requires scaling the results to the stand-level and situating them within the context of the regional timber supply and landscape occurrence of HDM. This gives shape to only one side of the comparisons described above. Finding ways to evaluate the impacts to forest components and processes affected by HDM infection and retention patches in the decision-making process must occur in tandem to completely describe the management decision.

4.1. Edge infection patterns

Our results show that HDM spread from infected mature forest edges ~35 years post-harvest is concentrated within 15 m and the severity is generally low. Incidence in live, dominant and codominant hemlock within 15 m of the mature forest edge averaged 69 % and DMR averaged 1.2. Impacts to tree height and diameter growth are assumed to begin at HDM infection levels of $DMR \geq 3$, which typically occur well into stand development (Thomson, Smith, and Alfaro 1984; Thomson et al. 1985; Muir and Hennon 2007). On average, 17% of dominant and codominant hemlock, which represent future crop trees, within 15 m of the edge, had passed this inflection point. No trees beyond 15 m had $DMR \geq 3$.

The modelling results formalize these observational statistics and conform with expected patterns of dwarf mistletoe epidemiology. The prediction from the model for dominant/codominant hemlock is that immediately adjacent to the edge (<10 m) DMR classes IBLC-2 and 3-5 are most probable; the probabilities of infected classes decline quickly with distance and beyond 15 m, the probability of DMR 3-5 is practically zero and DMR IBLC-2 is low (<20 %). The decrease in probability with increasing distance from the edge, at a rate that depends on seed load, is aligned with decreased exposure to seed sources and the dispersal distance of explosive seed discharge, ~ 15 m (Smith 1966; 1973; 1977; Parmeter 1978). Increasing probability of higher DMR classes with DBH has been observed repeatedly and is hypothesized to be driven by larger target size, longer exposure times and strong host vigour in large trees (Bickford, Kolb, and Geils 2005; Parmeter 1978; Shaw et al. 2005). The strength of the tree size effect is highest under overstory seed sources, where young small trees have traits that reduce the likelihood of infection (small target size, shorter exposure time, and poor vigour) and are shielded from overstory seed by larger tree layers (Parmeter 1978); this is represented in the model by the significant interaction of DBH and seed load. Each variable in the final model has a supporting mechanism in HDM epidemiology and this is the strongest evidence the “best” model was chosen in model selection. Complementing the AIC metric with the Bayesian information criterion (**BIC**) could have benefited model selection because BIC selects for the “correct” model, while AIC maximizes predictive accuracy (Aho, Derryberry, and Peterson 2014). This would have aligned with our inferential modelling objective.

Our results can be compared with the older set of field research measuring HDM infection around overtopping residual source trees that still serves as the evidence base for management today. Smith (1977) monitored open grown hemlock in southern coastal BC through a five-year exposure period to a single infected residual tree and a subsequent seven year period after the residual tree had been removed; at five years, incidence was 92% and after the subsequent seven years without the overhead seed source, tree DMR ranged from 0.2–2.3. This is a faster rate of spread and intensification than in comparable trees at our sites (dominant/codominant hemlock ≤ 15 m from edge), which had lower incidence and similar average DMR but are considerably older (23–45 years vs. 12 years). The stands at our sites were mostly

dense with closed canopies and non-hosts present. These conditions disfavour HDM (Richardson and van der Kamp 1972; Bloomberg and Smith 1982; Shaw and Weiss 2000) and contrast the open-grown conditions in Smith (1977). We expect this explains the differences in infection. Our results may better represent typical stands in coastal BC, because pre-commercial thinning is rare in current standard management and dense, closed-canopy conditions are the norm. Comparisons with studies in Alaska, support the observation that HDM development is slower in northern portions of HDM's range (Shaw III 1982; Shaw III and Hennon 1991; Trummer et al. 1998). The mechanisms for this effect have not been determined but are hypothesized to be related to climate and weather. For paired observations that cover similar timeframes, our results have somewhat higher rates of incidence and severity (Shaw III 1982; Shaw III and Hennon 1991).

The variation in the infection metrics and the magnitude of the random site effect in the modelling results indicate edge infection is variable across sites. Representative “high infection” sites have greater incidence of severely infected trees within 15 m of the edge and lightly infected trees distributed farther from the edge. “Low infection” sites are characterized by infrequent lightly infected trees that only occur near the edge. The obvious reason for this variation is that our sites differ in time since harvest. HDM infection increases with the length of exposure (Smith 1977; Shaw III and Hennon 1991) and there are indications of this in our data (e.g. mk_3, the youngest site [23 years], had minimal infection, all concentrated immediately adjacent to the edge). Other sites contradict this trend (e.g. ph_2, which is relatively old [40 years] and has almost no infection; cr_2 vs. cr_3, which are similar ages [38 and 37 years respectively] but have obvious differences in infection).

A host of other site variables affect HDM epidemiology and could explain the variation. Climate and weather regulate HDM infection through effects on plant development, seed survival and dispersal (Smith 1973; Barrett et al. 2012). The Pacific Highway cluster of sites represents the wettest climate zone in our data and consistently showed reduced infection. This may signal a climate effect. Winter is a critical period where seeds dispersed in the fall must remain attached to host branches to initiate infection the following spring and heavy rain can remove seeds (Barrett et al. 2012). This explanation is supported

by Pacific Highway sites having greater winter precipitation. The amount of seed being produced by the infection source and its relative position (i.e. the infection pressure) are also important predictors (Trummer et al. 1998; Robinson and Geils 2006). We represented this crudely for the mature component with the seed load variable and there is certainly uncaptured variation. The proxy does not consider dead infected hemlock source trees in the mature component, which may have contributed to infection pressure if they died recently. Only horizontal distances are considered, which ignores the influence of vertical position of infections in a crown and the relative position of trees to each other in the vertical plane. Infected amabilis fir are ignored as infection sources and were a major component of the ph_2 site. Lastly, the crown volume estimates that feed into the seed production proxy were limited by numerous assumptions (e.g. that the height ratio between crown classes is constant across sites).

Past research has identified numerous other site factors that are not incorporated in our model and lack a signal in our data (Parmeter 1978). Characteristics of the hemlock populations may be important; HDM shoot production is affected by host vigour (Bickford, Kolb, and Geils 2005) and early research has shown heritable resistant phenotypes in hemlock (Smith, Wass, and Meagher 1993). Stem density is a key variable discussed in the next section (Richardson and van der Kamp 1972; Bloomberg and Smith 1982; Shaw and Weiss 2000). The basal area of non-hosts is often considered because they act as barriers to spread (Bloomberg and Smith 1982). Numerous fungal pathogens that affect the shoots and endophytic system of dwarf mistletoes have been identified and may act to regulate HDM populations (Shamoun and DeWald 2002). All these factors are likely to vary across the sites, which is a product of our objective to cover a representative set of site conditions. Large variation between sites has been characteristic of HDM field research (Bloomberg and Smith 1982; Shaw III and Hennon 1991). For example, Shaw III and Hennon (1991) report 9–75% infection between plots of hemlock trees regenerating around infected residuals 25 years post-harvest.

4.2. Anticipated timber impacts at edge, stand and regional scales

HDM development to severities that cause growth impacts beyond the timepoint in our data will be functions of continued overhead seed exposure (for trees within 15 m of the edge), infection from neighbouring regenerating trees and reinfection from within the crown of a tree. Research from stands located in the same research forest as some of our sites has shown that hemlock height growth can match the rate HDM moves upwards within a crown; under this pattern trees largely escape HDM intensification and impacts to growth (Richardson and van der Kamp 1972). This pattern depends partly on site productivity (site index) and stem density, which control the rate of hemlock height growth, interception of seed and interior canopy conditions (e.g. light) that regulate HDM development (Richardson and van der Kamp 1972; Bloomberg and Smith 1982; Shaw and Weiss 2000). These factors could drive some differentiation in the rate of HDM development between sites but as a general prediction, we expect that trees without overhead seed exposure (>15 m) will escape or experience relatively small growth impacts over a typical rotation period where height growth remains high (e.g. 80 years). Growth impacts are expected in trees within 15 m of the edge that have existing infections. These trees face both continued overhead seed exposure to their upper crown and intensification within their lower crown from existing infections (Parmeter 1978). This reasoning is aligned Muir and Hennon (2007), who predicted growth impacts would be restricted to <10 m from infected edges; the extension from 10 to 15 m reflects common low-moderate infection in that interval in our data.

Edge infection patterns can be scaled to compare clearcut and group retention scenarios because these silvicultural systems differ in their edge to area ratio; in group retention, a larger proportion of the regenerating stand is growing near a mature forest edge, compared to clearcut. Hypothetical clearcut and group retention cutblocks are shown in Figure 12. Both assume a total area of 16 ha. The group retention block assumes $\sim 30\%$ retention (5.0 ha) distributed in regularly spaced 1.0 ha retention patches. These values are typical for coastal BC (Beese et al. 2019). Group retention criteria require that $\geq 50\%$ of the harvested area is within one tree length of a mature forest edge (BC Ministry of Forests 2003), which is

set at 35 m based on dominant/codominant mature trees in our data. Our prediction that growth impacts within a stand rotation will be limited to ≤ 15 m from the edge is used to define a second zone of edge influence. In the group retention scenario, the total harvested area is 11 ha, 8.4 ha (77%) is within a tree length of an edge and 5.2 ha (47%) is within 15 m of an edge. Relative to the clearcut, this adds 2.9 ha of area within 15 m of an edge and 0.5 ha of area that is within 15 m of two edges. If $\sim 30\%$ of the mature edge length (a number based on our field observations) has severe HDM infection, ~ 1.7 ha (15%) of the harvested area would be at risk of growth impacts; 1.0 ha of this infected area would be additional to a clearcut scenario (a $\sim 145\%$ increase). In the clearcut, 0.7 ha of the 16 ha harvested area (4%) is at risk of growth impacts. These figures account for areas under the influence of two edges by assuming they have equal and independent probability of experiencing infection from either edge.

Making these stand scale predictions requires various assumptions about the spatial arrangement of group retention stands and the levels of HDM infection in retained patches. The estimates above are speculative and represent a single static comparison but give context to the edge infection results. The large variation in patterns of edge infection will cause edge-affected areas to diverge in the time from harvest until severe infections occur and the proportion of trees infected. Though large, this variation is constrained to the area affected by HDM-infected edges, which in the hypothetical scenario, represents small to moderate proportions (4% and 15%) of the total harvested area in the clearcut and group retention cutblocks, respectively.

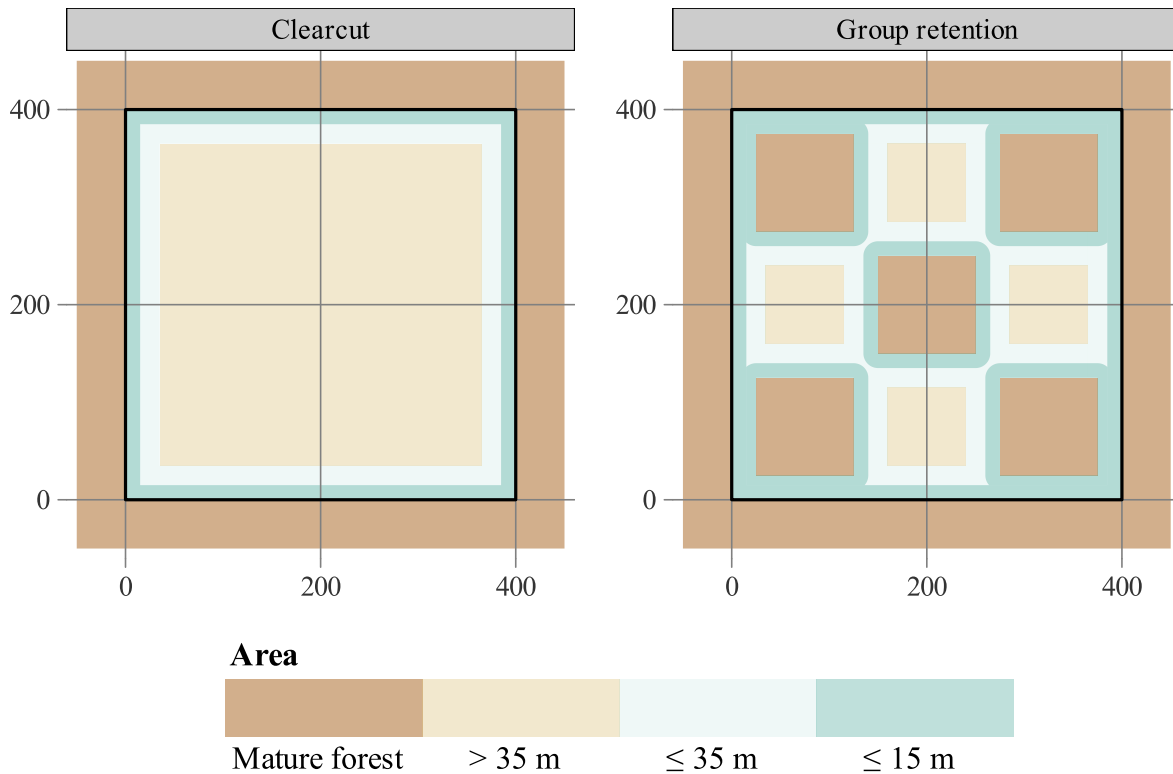


Figure 12. Hypothetical clearcut and group retention cutblocks showing zones of edge influence where HDM timber impacts might occur. The total cutblock area (black outline) is 16 ha, the average in coastal BC. Retention patches are 1.0 ha. Grid lines are in 200 m intervals. Numbers in legend refer to distance from a mature forest edge, located either at the cutblock boundary or in a retention patch. ≤ 35 m corresponds to a tree length. ≤ 15 m corresponds to the area where trees are at risk of growth impacts if an edge contains HDM infection.

The importance of these potential stand-level impacts depends on the regional context. This context defines the proportion of harvested volume and revenue that stand-level silvicultural system decisions affecting HDM and hemlock growth, have the potential to influence. In our case study, hemlock accounted for $\sim 15\%$ of the future harvested volume and $\sim 5\%$ of the anticipated revenue (represented by VCP). This mismatch between hemlock's contribution to future volume and revenue creation reflects the influence of Douglas-fir and cedar, which were major contributors to future volume and are higher-valued.

These data points are informed by landscape occurrence estimates of HDM, which are only available from summaries of forest inventory data in the western United States (Bolsinger 1978; Hildebrand 1995; Barrett et al. 2012). Forest inventory data are limited by challenges with field crews not

recognizing subtle dwarf mistletoe symptoms and stratification that can introduce bias (Muir and Moody 2002) but provide sufficient precision for discussion. For the most recent period summarized (1995–2003), 14% of plots with hemlock trees had HDM present and 5% of all hemlock trees were estimated to be infected, across Washington, California and Oregon; results were similar for Alaska (Barrett et al. 2012). The geographic range of hemlock is greater than that of HDM, so these occurrence estimates are representative of a host range that HDM occupies only a part of, and they do not necessarily reflect the BC context. In regions with a larger overlap between hemlock and HDM ranges, occurrence is likely higher (see examples in Hildebrand 1995).

Acknowledging the limitations in the occurrence estimate, and that the timber supply figures come from a single case study, their combined conclusion is that there is a relatively constrained space for stand-level decisions affecting HDM levels to translate to impacts on timber supply and revenue. HDM is likely to be a decision-making factor in ~15% of stands (depending on the region) and its potential impact on revenue from timber products is limited by other timber species (cedar and Douglas-fir) that are typically preferred in silviculture planting prescriptions and higher-valued.

4.3. Ecological functions of HDM and retention patches

Evaluating the impact of forest management decisions is challenging because of the number and complexity of forest components and processes involved. In the context of forest diseases, impact assessments have typically focused on timber resources, in part because they are relatively easy to measure (Lundquist, Goheen, and Goheen 2002). The terms *HDM ecological functions* and *retention ecological functions* are used here to refer to the various forest components (e.g. individual species, stand structure, wildlife habitat) and processes (e.g. succession, nutrient cycles, resilience to disturbance) that HDM and retention patches have core influences on. Many of these functions are not easily quantified and even when subsets of interest can be measured, weighing their importance against other factors in decision-making, is context dependent and relies on their relationship to defined management objectives (Lundquist, Goheen, and Goheen 2002). Despite these challenges, they are factors that forest managers

have to regularly incorporate into their decision making. Characterizing trade-offs, synergies and null relationships between impacts to these functions and timber resources is part of selecting silvicultural systems in stands with HDM. In a hypothetical stand where HDM and variable retention objectives are the primary decision-making factors, selecting a clearcut silvicultural system prioritizes timber production. Selecting group retention prioritizes HDM and retention ecological functions. This is not a binary decision in practice. Intermediate options are possible (e.g. group retention that intentionally selects healthy trees along retention patch borders) (Rusch et al. 2019), but it provides a general frame for discussing what is prioritized at each extreme.

Retention patches are stand scale elements of the larger landscape-level variable retention approach (BC Ministry of Forests 2003). Retention ecological functions that occur at the stand scale (e.g. refugia for species specific to mature forests to persist over a rotation) are linked to functions that occur at the landscape scale (e.g. total habitat for species specific to mature forests) (Beese et al. 2019; Carrol 2023). This spatial dependence implies that in any given stand, the impact of an HDM-based silvicultural system decision on retention ecological functions will depend upon the existing patterns of mature forest in the surrounding landscape. Similarly, whether default use of clearcut systems in stands with HDM will affect landscape-scale retention functions depends on whether HDM occurs frequently enough to compromise landscape variable retention objectives. Relevant too is that retention patches have a timber impact. In the hypothetical comparison discussed above (Figure 12), retention of 30% of the mature forest for a full stand rotation (a retention system criterion) (BC Ministry of Forests 2003) is a large reduction relative to a scenario that maximizes yield. Balancing the retention of ecological functions with timber resources, which are impacted by both retention patches and HDM, requires planning beyond stand-level decisions.

The relationship between HDM ecological functions and HDM associated timber impact contains trade-offs. Lundquist, Goheen and Goheen (2002) measured the impact of Douglas-fir dwarf mistletoe infection on timber production and structural diversity and showed that stands representing each objective had light and severe infection ratings, respectively. This likely generalizes to the subset of HDM

ecological functions that occur through the same processes that cause timber impacts. For example, the association of marbled murrelets and stands with HDM infection (Hamer et al. 2008) is hypothesized to be a habitat selection preference for open, structurally complex canopies. HDM generates structural complexity and stand openness through crown transformations and single tree mortality, which also drive its timber impact (Meinzer, Woodruff, and Shaw 2004; Marias et al. 2014; Bell, Pabst, and Shaw 2020; Calkins, Shaw, and Lan 2021). We expect this trade-off has a convex (stable) shape and does not begin until stands experience moderate severity infection. When moderate severity infection levels are reached, benefits to structural complexity and stand openness will accrue at incremental losses to timber production.

Weighing the trade-offs between HDM ecological functions and timber production in the context of group retention silvicultural systems is informed by the “healthy level of disease” concept (Manion 2003); native forest pathogens, occurring within their natural range of variability contribute to forest health (Manion 2003; Ostry and Laflamme 2009; Carrol 2023). Forest health is defined by a reference condition in a forest’s ecological processes and resistance and resilience to disturbance (Carrol 2023); it is not necessarily aligned with the objectives areas managed for timber production (Raffa et al. 2009; Wingfield et al. 2015; Carrol 2023). However, it is at least partly aligned with the objectives of areas managed under the variable retention approach, because the approach aims to balance timber production and ecological function with an underlying logic of replicating natural processes of disturbance (Beese et al. 2019; Kuuluvainen et al. 2021).

The endemic levels of HDM infection that occur in unmanaged forests subject to a natural disturbance regime are a candidate measure for defining the natural range of variability that corresponds to a healthy level of disease. This assumes HDM is regulated in unmanaged forests through its interaction with other components of a disturbance regime (e.g. wind, fire, gap-disturbance) (Shaw et al. 2005; Shaw and Agne 2017; Swanson, Shaw, and Marosi 2006), resistance in potential hosts, diseases (Shamoun and DeWald 2002) and other factors. This is a species-scale adaption of the assumption that is applied at the landscape-scale to characterize healthy levels of disturbance; in unmanaged landscapes without outside

pressures (e.g. climate change), disturbance regimes (and the agents that drive them) are assumed to operate within a stable range, that gives rise to a diverse, self-sustaining, healthy forest landscape (Carroll 2023). This assumption has not been tested for HDM nor have infection levels in unmanaged landscapes been measured. Initial research has documented the interaction of fire and dispersal on HDM distributions (Shaw et al. 2005; Swanson, Shaw, and Marosi 2006; Shaw and Agne 2017). This has contributed to an understanding of HDM's role and interaction with different disturbance regimes in the Pacific Northwest (Muir and Hennon 2007) but the endemic levels of infection that might result remain conceptual predictions.

This lack of data does not necessarily limit applying the reasoning of the healthy level of disease concept. The fundamental assumption of variable retention is that by replicating patterns of natural disturbance, associated species, communities, processes and services are maintained (Beese et al. 2019; Kuuluvainen et al. 2021). This logic is applied to all species that lack fine-resolution data from studies in retention stands and is extension of the coarse-filter principle that originated in conservation planning (Lemelin and Darveau 2006). In the CWH ecosystem in coastal BC, group retention silvicultural systems attempt to replicate patterns in forest structure that arise from a disturbance regime dominated by small-scale gap disturbance (BC Ministry of Forests and Ministry of Environment, Lands and Parks 1995; Daniels and Gray 2006). Following the coarse-scale principle, levels of HDM in group retention stands are expected to fall within their natural range of variability because harvesting resembles gap disturbance. For forest managers, an alternative, practical argument for intentionally incorporating HDM in group retention stands is simply that HDM ecological functions contribute to non-timber objectives (e.g. wildlife habitat and structural diversity) (Beese et al. 2019).

4.4. Limitations

There are limitations in the study design, field methods and modelling portions of this research. The edge infection field sites have different management histories and at least three had juvenile spacing treatments. Hemlock is often disfavoured for higher timber value species in our region, and it is possible

it would have been selectively removed for this reason, or even removed because of HDM infection. We assume that the mature component was the sole source of HDM at each site, which ignores rare, animal mediated dispersal (Ostry, Nicholls, and French 1983; Nicholls, Egeland, and Hawksworth 1989) and infected residual trees potentially left during harvesting. Management guidelines have been in place since 1972 recommending the removal of residual trees (Baranyay and Smith 1972; Rusch et al. 2019) but historically, management that left these trees has been the primary silvicultural concern for HDM. Ten trees were identified in the regenerating component across the sites that had signs of being infected pre-harvest. All were close to the edge and resembled the dominant/codominant tree layer tree size but were not true, overtopping residuals. We considered them infected advance regeneration, removed them from the modelling dataset and interpret their influence as source trees to be small.

The visual assessment of HDM infection with the six-class DMR system (Hawksworth 1977) is known to be inaccurate in dense coastal forests (Muir and Hennon 2007). Shaw et al. (2000) compared ratings of ground observers to those from a canopy crane in an old-growth hemlock-Douglas-fir forest and found observers were good at assessing incidence (73–95% correct) but poor at assessing DMR (21–58% of trees incorrectly rated by ≥ 2 DMR classes). We experimented with unmanned aerial vehicles when developing our methods but found they helped little and interpret the ground observer bias as a necessary source of error.

Using simulated data for modelling is likely a source of bias that favours underpredicting DMR. It is possible that live hemlock trees beyond the end of transects at our sites (which are assumed to be healthy in simulated data) were infected; this would bias predictions towards lower DMR. More problematic, we expect that hemlock declines in abundance towards the interior portions of harvested areas because it is not planted and seeds naturally. Hemlock seed abundance declines exponentially with distance from the source, and most is dispersed within 500 m (Harris 1967; Klinka et al. 2000). Simulated data was based on composition data from close to the edge (from 5 m to the transect ends at each site) and this likely leads to hemlock being overrepresented. The most extreme case are sites with 15 m transects; at these sites, composition data is derived from 5–15 m on each transect (immediately next to mature

hemlock seed sources) and used to simulate data from 15–50 m. In effect, this inflates the number of healthy hemlocks in the modelling dataset and biases predictions towards lower DMR classes. Over the scale of ~50 m, we expect hemlock abundance remains relatively high, and this bias is small. In contrast, not simulating data, would have biased predictions towards higher DMR classes. Transects were extended beyond 15 m to capture sites with greater spread; if models were built on field measured data without standardizing transect length, infected trees would be overrepresented in observations from 15–50 m.

Chapter 5: Management implications and future research

The tentative management implication from this research is that the potential timber revenue impact from HDM in group retention over standard rotations in coastal BC is relatively small and may not justify use as a decisive rationale for silvicultural system prescriptions. Part of this conclusion is based on anticipated impacts to hemlock growth being restricted to areas within 15 m of infected edges. A second part comes from the constraints of the regional context. Hemlock is not the highest valued species in coastal BC and HDM is estimated to occur in only a portion of the stands with hemlock (Hildebrand 1995; Barrett et al. 2012; Bolsinger 1978). The conclusion is also supported by the idea that forest areas managed for both timber and ecological objectives, with an underlying natural disturbance-based strategy, should intentionally incorporate native disturbance agents that contribute to forest health (Manion 2003; Ostry and Laflamme 2009; Carrol 2023). However, the results presented here are preliminary, and the data used to describe the regional context is sparse or relies on assumptions. A structured set of future research, organized by priority, is recommended before formal updates to HDM management guidelines occur (Rusch et al. 2019).

5.1. Priority 1: Problem characterization with existing data

Recommended research

1. Simulating stand-level infection at ~35 years post-harvest with the edge infection equations developed in this research using spatial data from actual group retention stands.
2. Adding additional licensee areas to the technoeconomic analysis to better characterize the regional importance of hemlock to harvested volume and revenue.
3. Summarizing HDM occurrence data with forest inventory data from BC. The Change Monitoring Inventory program data contains the necessary data for this task (Forest Analysis and Inventory Branch 2024a).

Together this research would likely give sufficient estimates of the potential timber-impacts involved in the management problem to inform updated guidelines and investment in future research.

5.2. Priority 2: Continued research

Recommended research

1. Continued monitoring of the edge infection research sites to track HDM infection with stand development. Five-year intervals are recommended (Hawksworth and Marsden 1990).
2. Integrating the existing Dwarf Mistletoe Impact Model (**DMIM**) (Robinson and Geils 2006) with a spatially explicit growth and yield model capable of simulating group retention stand development.
3. Estimating endemic levels of HDM infection in unmanaged stands and landscapes subject to a natural disturbance regime.

Remeasurements of the edge infection research sites will provide basic data about HDM development in edge effected areas to severities associated with timber impacts. The future timber impacts discussed here rely on assumptions about HDM development. Collecting data over a full rotation would remove the need for assumptions and inform management guidelines irrespective of the magnitude of the timber impact the data shows. In addition, with multiple measurements, the time since harvest variable would be within a site. This would remove its confounding effect on between site comparisons and allow for targeted studies on variables that differ between sites and affect HDM biology (e.g. climate, condition and genetic makeup of hemlock populations) (Smith, Wass, and Meagher 1993; Bickford, Kolb, and Geils 2005; Barrett et al. 2012).

If research in Priority 1 indicates potential HDM timber impacts in group retention are large enough in some contexts to warrant the development of stand-level decision making tools to support forest managers, a modelling based approach could complement field studies. The DMIM is a mechanistic model of dwarf mistletoe spread and intensification (Robinson and Geils 2006). It is integrated within the Forest Vegetation Simulator (**FVS**), which is the primary growth and yield model

for managed forests in the United States (Dixon 2024). The DMIM is limited for the group retention use case by the model architecture of FVS. In FVS, trees are represented by statistical spatial distributions and stands with non-uniform horizontal distributions of trees in different size classes cannot be generated (Penner 2021). The representations of mistletoe processes in the DMIM (e.g. seed dispersal) are transferable to other growth and yield models. Possible spatially explicit models that could integrate the DMIM include the Tree and Stand Simulator (**TASS**) and the SORTIE model (Canham and Murphy, n.d.). This represents a large research effort that is limited by models being closed access (TASS), using outdated programming languages (the FORTRAN language in FVS), not being developed for complex stand structures (FVS and TASS) and not being calibrated for coastal BC (SORTIE) (Penner 2021). However, this approach would expedite results relative to waiting for data from field sites and develop the framework for other dwarf mistletoe-host systems.

Endemic levels of HDM in unmanaged stands and landscapes would provide a reference condition to interpret HDM levels in managed forests against. This research can build from existing research on fire's influence on HDM distributions (Shaw et al. 2005; Swanson, Shaw, and Marosi 2006; Shaw and Agne 2017) and explore HDM's interaction with other disturbance agents and factors that regulate its population. The objective of this research would be to provide quantitative estimates of expected infection incidence, severity and spatial patterns at different scales and under different disturbance regimes. Beyond its management utility, this research could test the evolutionary question of whether a stable, endemic level, of HDM infection exists; genetic and genomic methods would be valuable here because of their capacity to answer questions in historical ecology.

Appendix

Table A1. Site selection criteria. R = regenerating component, M = mature component, Hw =hemlock, HDM = hemlock dwarf mistletoe.

Criteria	Required value	Rationale
Leading tree age	R: 20–45 years M: >100years	20–45 years was chosen because: (1) preliminary studies in our region found HDM infection does not appear until >20 years post-harvest (Muir 2010) and (2) management recommendations to remove infected residual trees were in place by 1972, so sites in this range are not likely to include infected residuals and edge effects can be isolated (Baranyay and Smith 1972). An age differential of >50 years ensures overtopping edge trees constitute a significant overhead seed source.
Species composition	R: Hw ≥30% M: Hw ≥30%	High Hw composition means site is at risk of HDM infection. Sites with secondary hosts of HDM were avoided to create a comparable sample of sites where Hw was the primary host controlling HDM dynamics.
Biogeoclimatic zone	Coastal Western Hemlock (CWH)	All CWH subzones are rated high hazard for HDM (Rusch et al. 2019); site selection was distributed across the precipitation range in the CWH to identify related trends.
Logging disturbance	One pass	Ensures consistency among sites.

Table A2. Climate and physical characteristics of the study sites. BEC = biogeoclimatic zone to subzone or variant level, SI = site index for hemlock, MAT = mean annual temperature, Tmin_{cm} = mean minimum daily temperature of the coldest month, MAP = mean annual precipitation, PAR = annual precipitation as rain, PAS = annual precipitation as snow. Cluster: CR = Campbell River, FV = Fraser Valley, PH = Pacific Highway.

Site	Cluster	BEC	SI (m)	Elevation (m)	Aspect (°)	Slope (°)	MAT (°C)	Tmin _{cm} (°C)	MAP (mm)	PAR (mm)	PAS (mm)
cr_1	CR	CWHxm2	24.0	95	88	-13	9.4	1.1	1772	1690	82
cr_2	CR	CWHxm2	24.0	137	89	-4	9.1	0.8	2021	1916	105
cr_3	CR	CWHxm2	24.0	85	16	-8	9.4	1.1	2084	1991	93
mi_1	FV	CWHdm-	30.3	196	112	-14	9.9	0.6	2274	2178	96
mi_2	FV	CWHdm-	27.2	199	237	-13	9.9	0.7	2489	2385	104
mk_1	FV	CWHdm-	28.2	247	110	-13	9.6	0.5	2245	2138	107
mk_2	FV	CWHvm2	27.0	391	23	-27	9.0	-0.1	2420	2280	140
mk_3	FV	CWHdm-	28.3	195	118	-14	9.8	0.6	2170	2074	96
ph_1	PH	CWHvm1	27.9	267	230	-38	8.7	0.4	4007	3803	204
ph_2	PH	CWHvm1	32.6	69	-	0	9.5	1.2	3944	3810	134
ph_3	PH	CWHvm1	17.2	59	76	-19	9.6	1.7	3372	3275	97

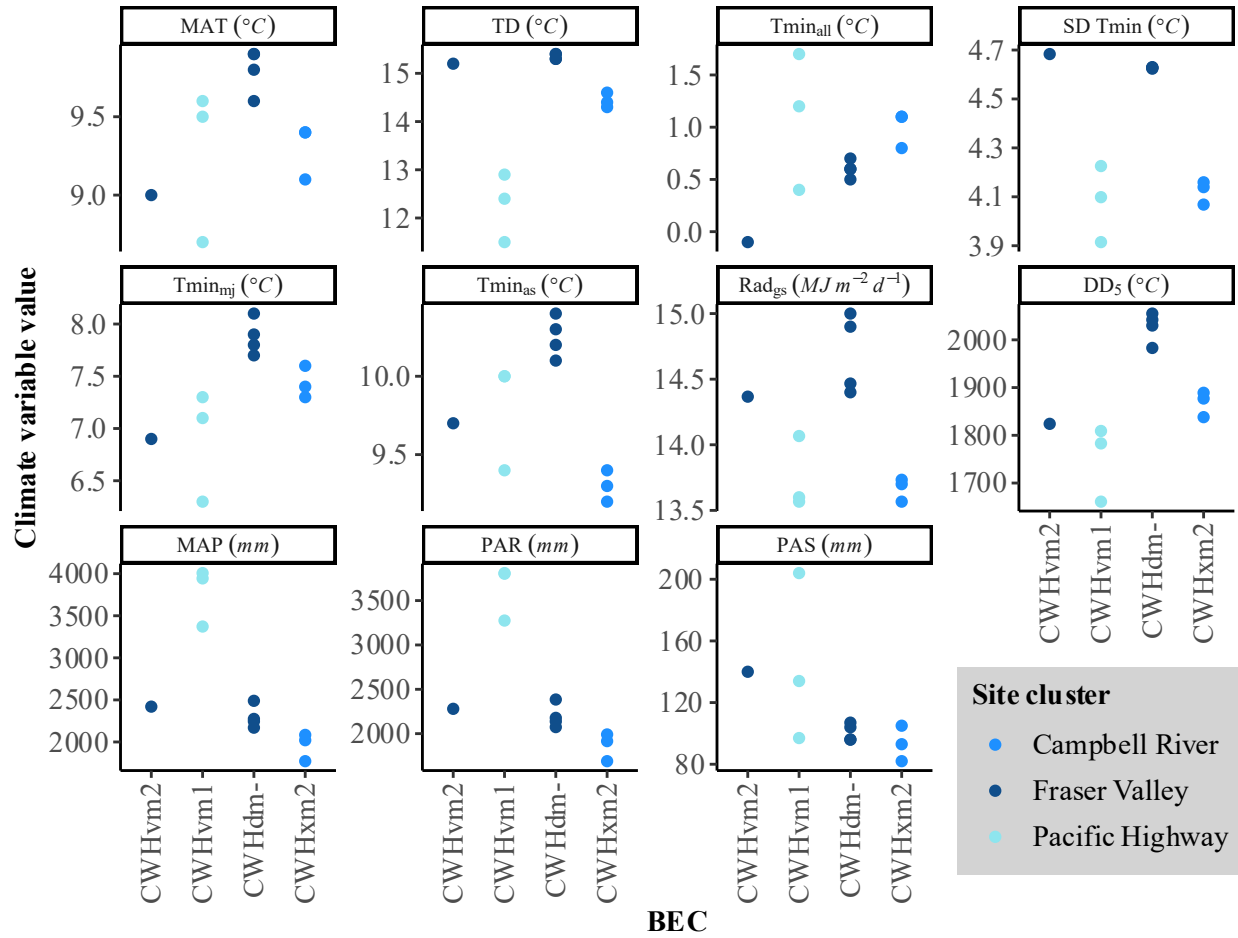


Figure A1. Climate profiles of the study sites. Raw data are from ClimateBC (v7.50) for the Normal 1991–2020 period. MAT = mean annual temperature, TD = temperature differential (difference between mean warmest month temperature and mean coldest month temperature), Tmin_{all} = lowest average minimum monthly temperature of all months, SD Tmin = standard deviation of lowest average minimum monthly temperature across all months, Tmin_{mj}/Tmin_{as} = lowest average minimum monthly temperature between May and June and August and September, respectively, Rad_{gs} = growing season radians, DD₅ = degree days above 5 $^{\circ}\text{C}$, MAP = mean annual precipitation, PAR = precipitation as rain, PAS = precipitation as snow.

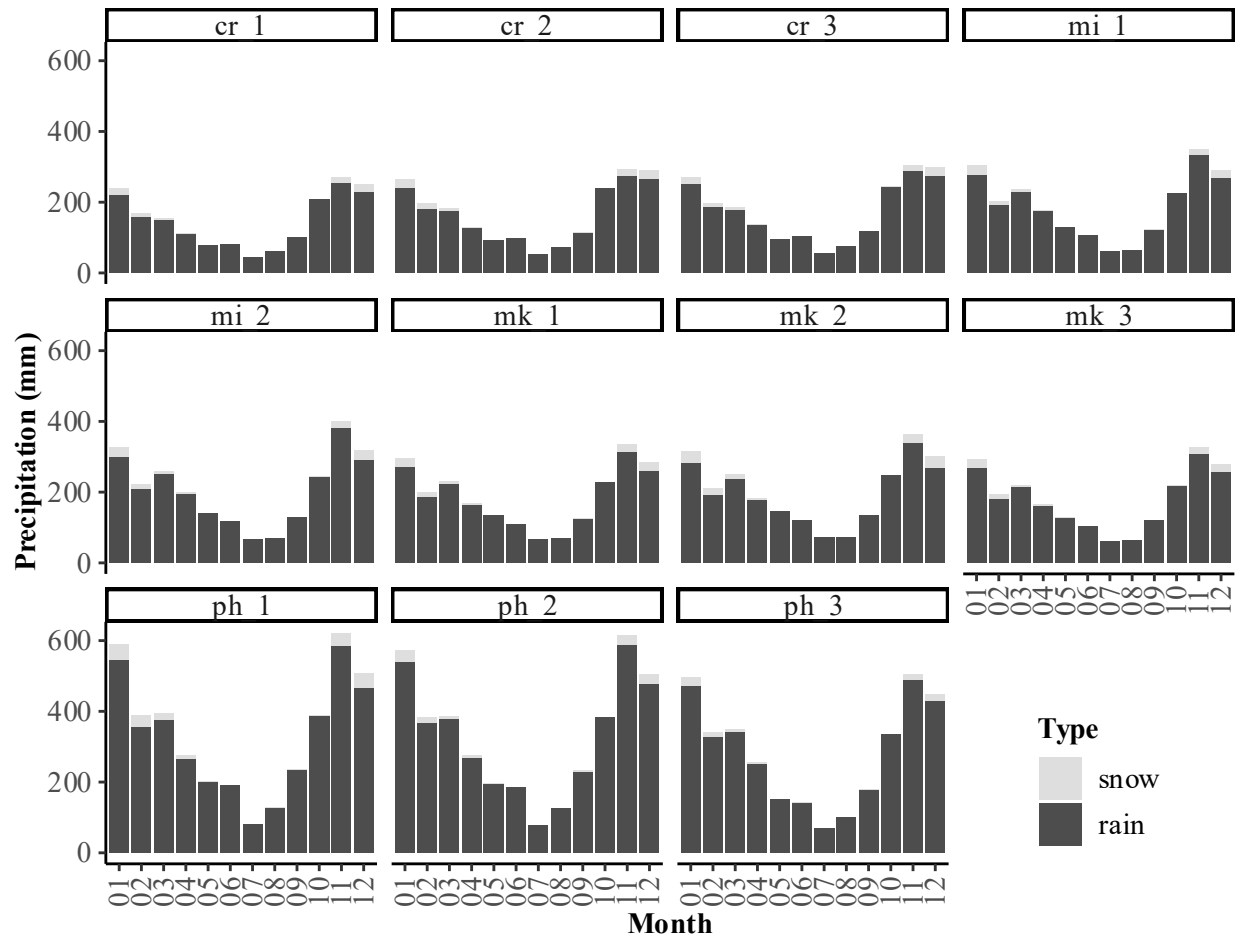


Figure A2. Monthly precipitation (mm) as snow and rain at each study site. Raw data are from ClimateBC (v7.50) for the Normal 1991–2020 period.

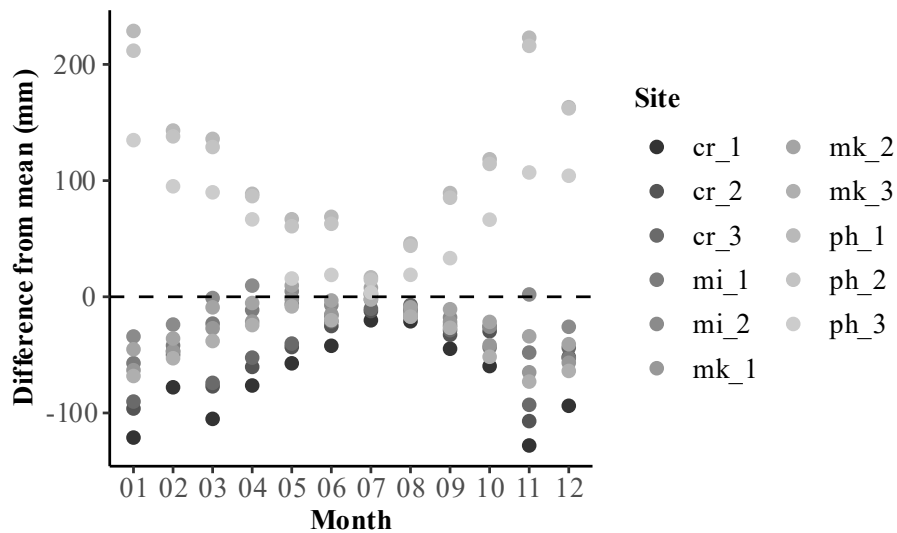


Figure A3. Difference between site precipitation and mean precipitation across sites for each month (mm). Raw data are from ClimateBC (v7.50) for the Normal 1991–2020 period.

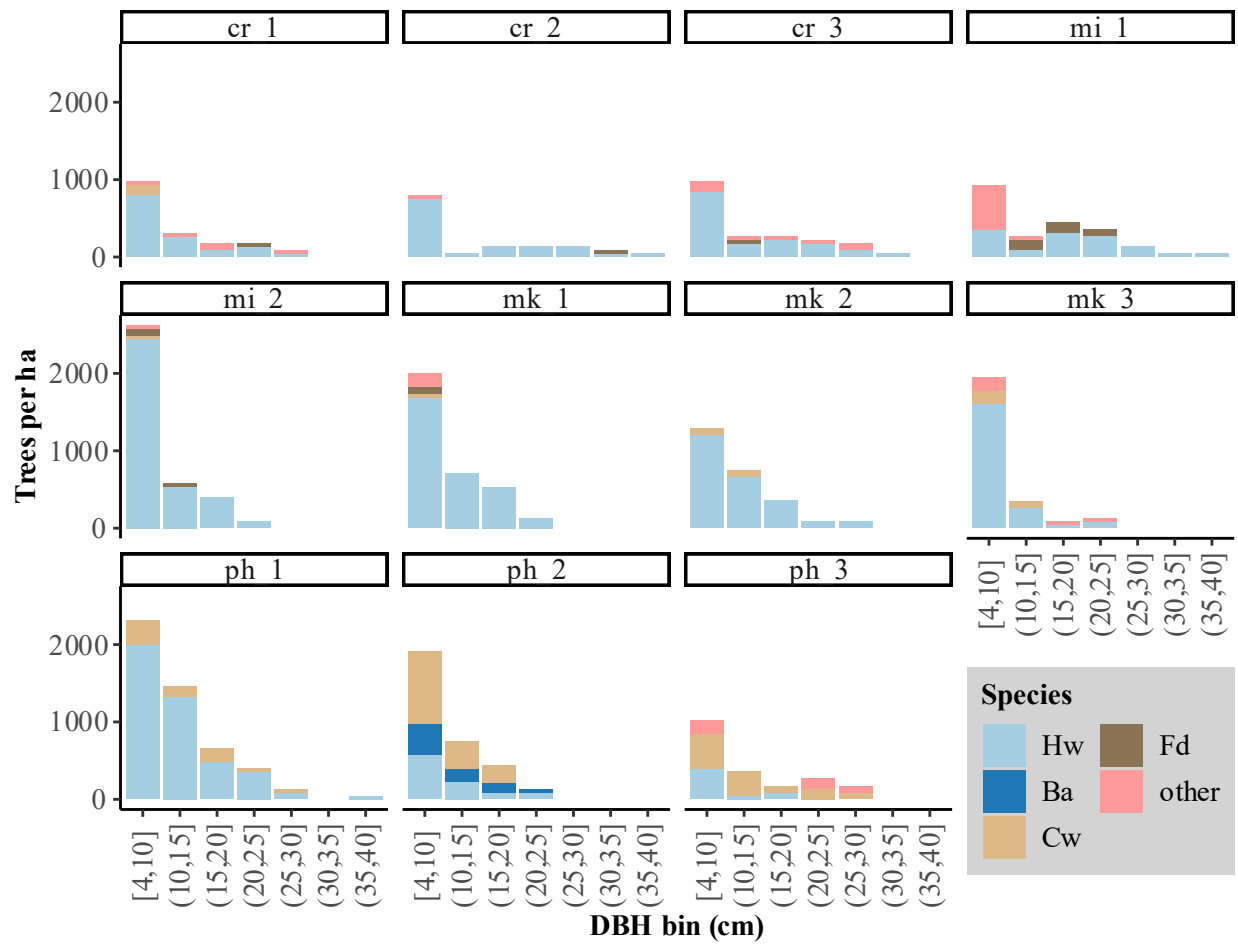


Figure A4. Size class distribution and composition of regenerating forest components across the study sites. Data are from the 0.0–15.0 m interval of the three transects at each site and include live and dead trees ≥ 4.0 cm. Species: Hw = hemlock, Ba = amabilis fir, Cw = cedar, Fd = Douglas-fir.

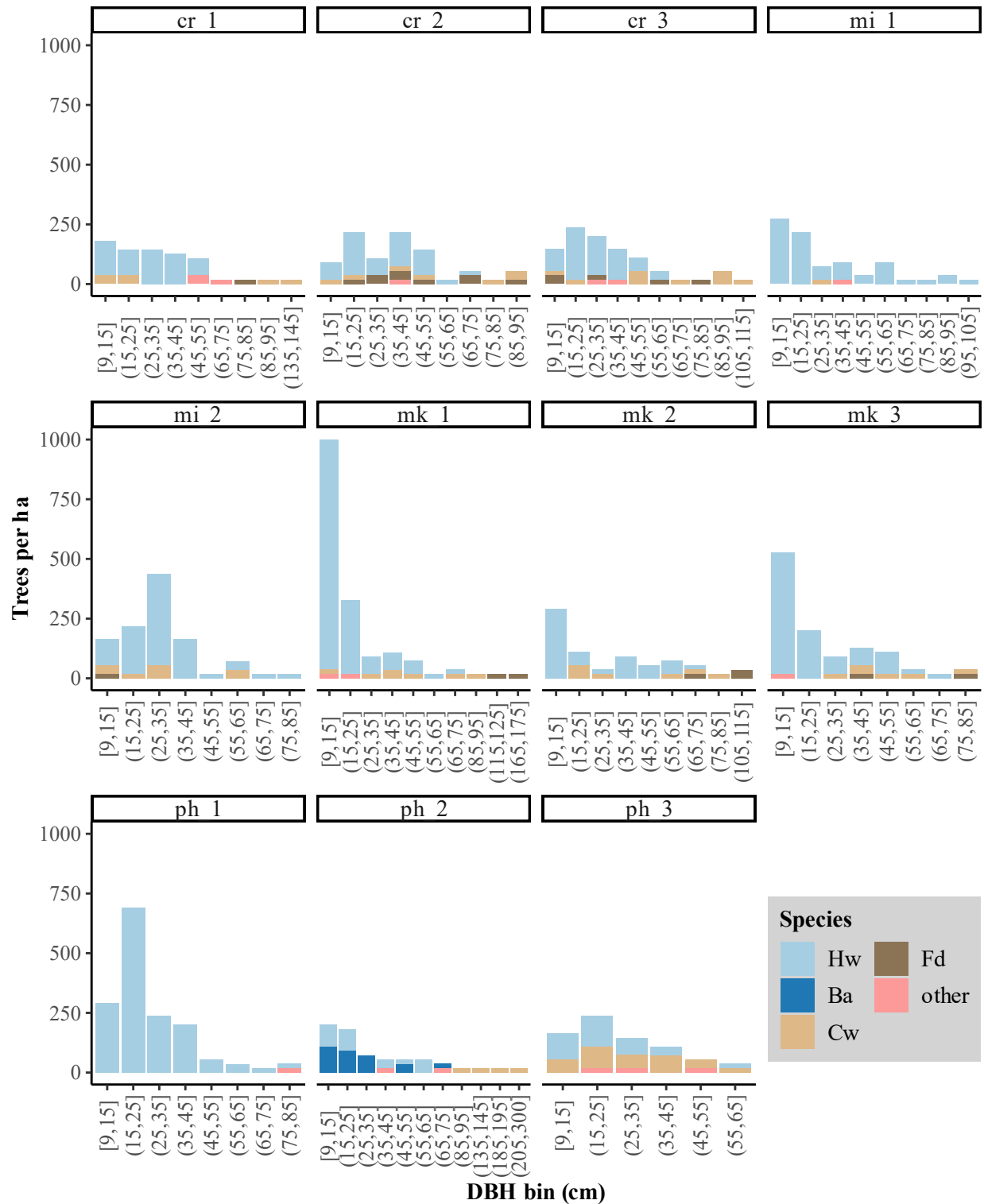


Figure A5. Size class distribution and composition of mature forest components across the study sites. Data are from the 10.0 x 55.0 m survey area at each site and include live and dead trees ≥ 9.0 cm. Species: Hw = hemlock, Ba = amabilis fir, Cw = cedar, Fd = Douglas-fir. Note that x-axis scale differs between panels.

Table A3. Site history of regenerating forest components. All regenerating components are in larger areas that were harvested as clearcuts in the year indicated. Area harvested indicates total size of harvested area. Stocking standards indicate the species desired by the forest managers; P = preferred, A = acceptable, spacing = target density of well-spaced trees. Planting and treatment columns indicate events (e.g. tree planting), the year they occurred, their area (compare with total harvest area) and, for planting only, the densities of planted trees (s/ha = stems/hectare). Codes are used to indicate species: Hw = hemlock, Fd = Douglas-fir, Cw = cedar, Bg = *Abies grandis*, Ba = amabilis fir, Ss = *Picea sitchensis*, Ax = *Populus* hybrid, Dr = *Alnus rubra*. Site history is missing for some sites (indicated with '-'); for sites where it is available, note that planting and treatments often apply to only a portion of the harvested area and may not overlap with the area covered by our research site.

Site	Year harvested	Area (ha)	Stocking standards	Planting	Treatments
cr_1	1990	15	P: Fd, Cw, Bg A: none Spacing: 900 s/ha	1992 (14 ha): Fd (607 s/ha), Cw (129 s/ha)	Spot burned (1 ha, 1991)
			P: Fd, Cw A: none Spacing: unknown	1987 (14 ha): Fd (927 s/ha)	Manual weeding (3 ha, 1992) Juvenile spacing, Fd and Cw favoured (15 ha, 1998)
cr_2	1985	15	P: Fd, Cw, Hw A: Bg Spacing: unknown	1989 (15 ha): Fd (943 s/ha), Cw (300 s/ha)	Chemical herbicide (9 ha, 1993) Manual brushing (2 ha, 1994; 7.3 ha 1997)
			P: Fd, Cw, Hw A: Act Spacing: 900 s/ha	1996 (11 ha): Fd (626 s/ha), Cw (231 s/ha), Ax (48 s/ha)	Manual brushing (1 ha, 1996; 4 ha, 1999; 1 ha, 2000; 10.5 ha, 2002)
cr_3	1987	16	-	-	Juvenile spacing (details unknown)
mi_1	1991	7	-	-	-
mi_2	1994	13	P: Fd, Cw, Hw A: Act Spacing: 900 s/ha	1994 (1 ha): Fd (320 s/ha), Cw (360 s/ha) 1995 (5 ha): Fd (642 s/ha), Cw (397 s/ha)	Manual brushing (1 ha, 1996; 4 ha, 1999; 1 ha, 2000; 10.5 ha, 2002)
			-	-	-
mk_1	1990	10	-	-	-
mk_2	1978	4	-	-	-
mk_3	2000	8	-	-	-
ph_1	1983	71	P: Hw, Ba A: none Spacing: unknown	1984 (21 ha): Ss (857 s/ha) 1985 (46 ha): Ba (978 s/ha) 1987 (2 ha): Cw (1000 s/ha) 1988 (46 ha): Ba (326 s/ha)	Broadcast burn (21 ha, 1983; 46 ha, 1984) Juvenile spacing (area unknown, 1998) Manual brushing (36 ha, 2001) Fertilization (53 ha, 2004)
			P: Cw, Ba A: Ss, Hw Spacing: 900 s/ha	1991 (18 ha): Cw (773 s/ha), Ba (511 s/ha), Ss (104 s/ha) 1994 (13 ha): Cw (fill planted)	Manual brushing (10ha, 1993; 2.3 ha, 2001)
ph_2	1989	20	P: Cw, Ba A: Hw Spacing: 900 s/ha	1993 (25 ha): Cw (732 s/ha), Ba (203 s/ha)	Pile and burn (25 ha, 1993) Manual brushing, targeting Dr (10 ha, 2001)
ph_3	1991	28	-	-	-

Table A4. Observers present at each study site.

Observer	Sites surveyed
1	All
2	cr_1, cr_2, mi_1, mi_2, mk_1, mk_2, mk_3, ph_1
3	mk_1, cr_3
4	ph_2, ph_3

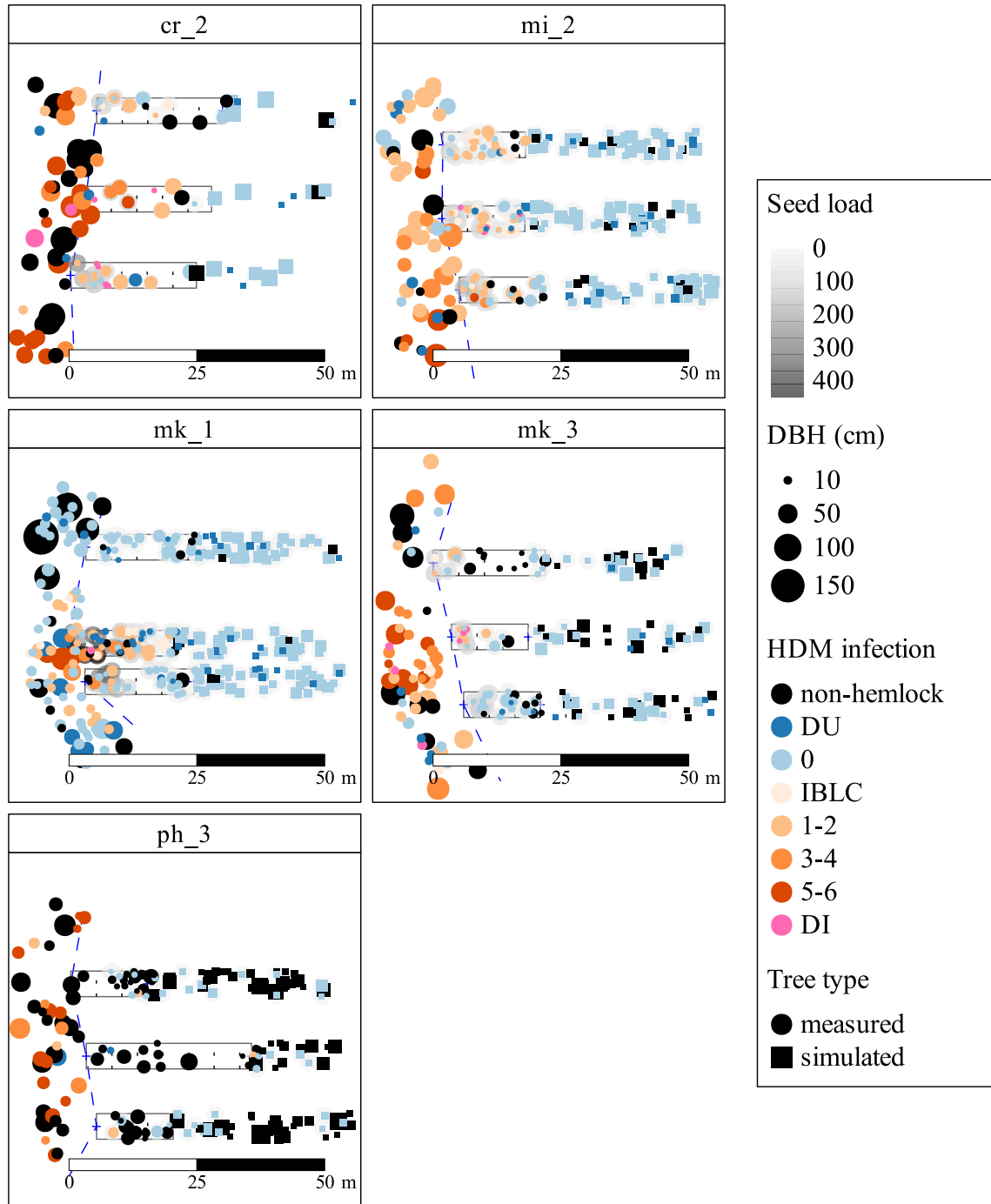


Figure A6. Stem maps showing hemlock dwarf mistletoe (HDM) infection at five sites omitted in main text figure. Mature forest component is on left side in each panel. Regenerating component is on the right. Live hemlock trees in regenerating component are overlaid on a background of seed load – a proxy for the amount of dwarf mistletoe seed deposited on each tree from infected hemlock in the mature component. For HDM infection, dead trees are recorded on a presence/absence scale (DU = dead uninfected, DI = dead infected) and live trees are recorded on the dwarf mistletoe rating (DMR) scale (0, IBLC = infected below live crown, 1–6).

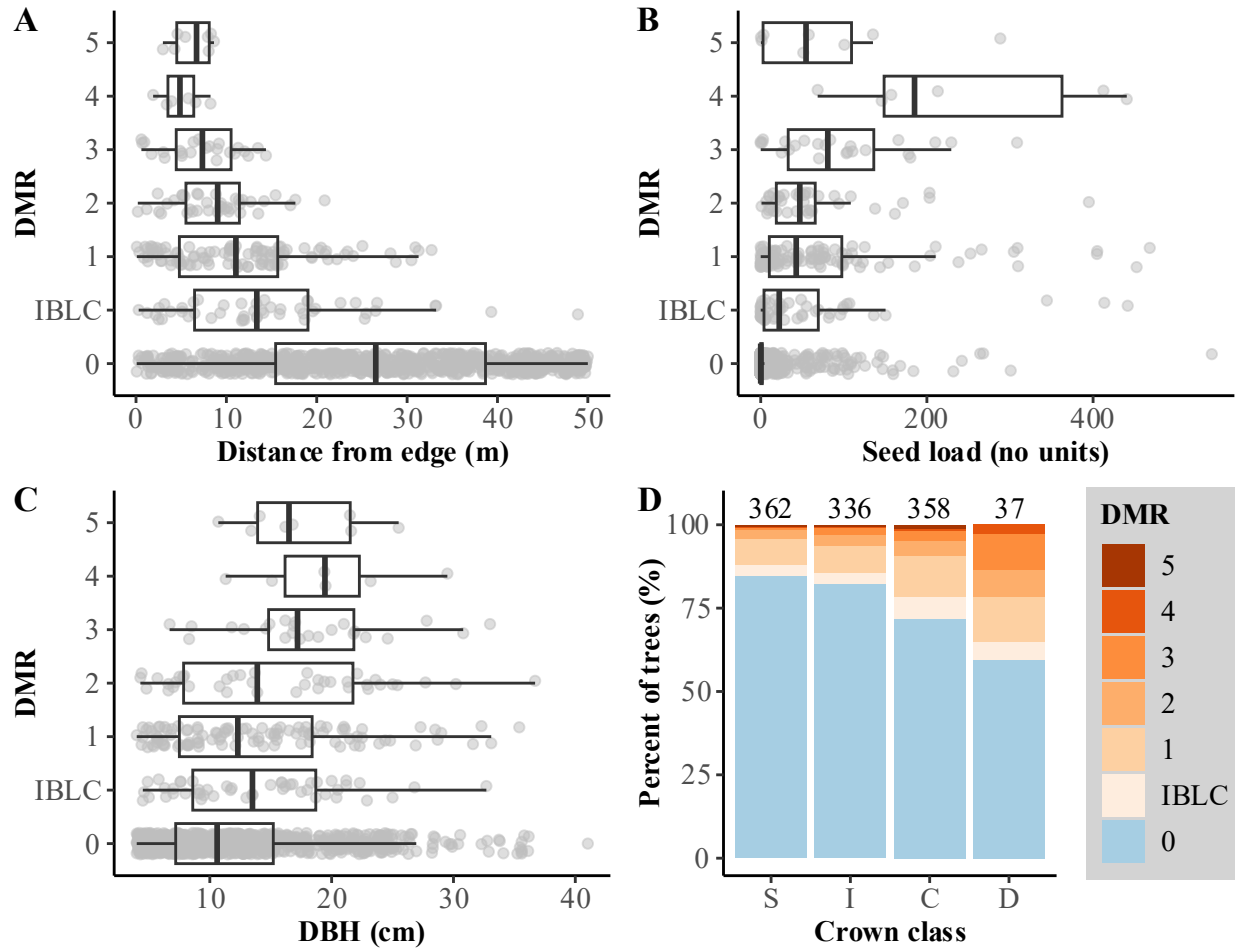


Figure A7. Dwarf mistletoe rating (DMR) plotted against tree-level predictor variables. Data are the modelling dataset—live hemlock trees in the regenerating forest component from 0.0–50.0 m from the edge. In (D), numbers at top of bars are number of trees. IBLC = infected below live crown. In boxplots (A–C), upper and lower edges of boxes are 75th and 25th percentiles; points beyond whiskers are outliers.

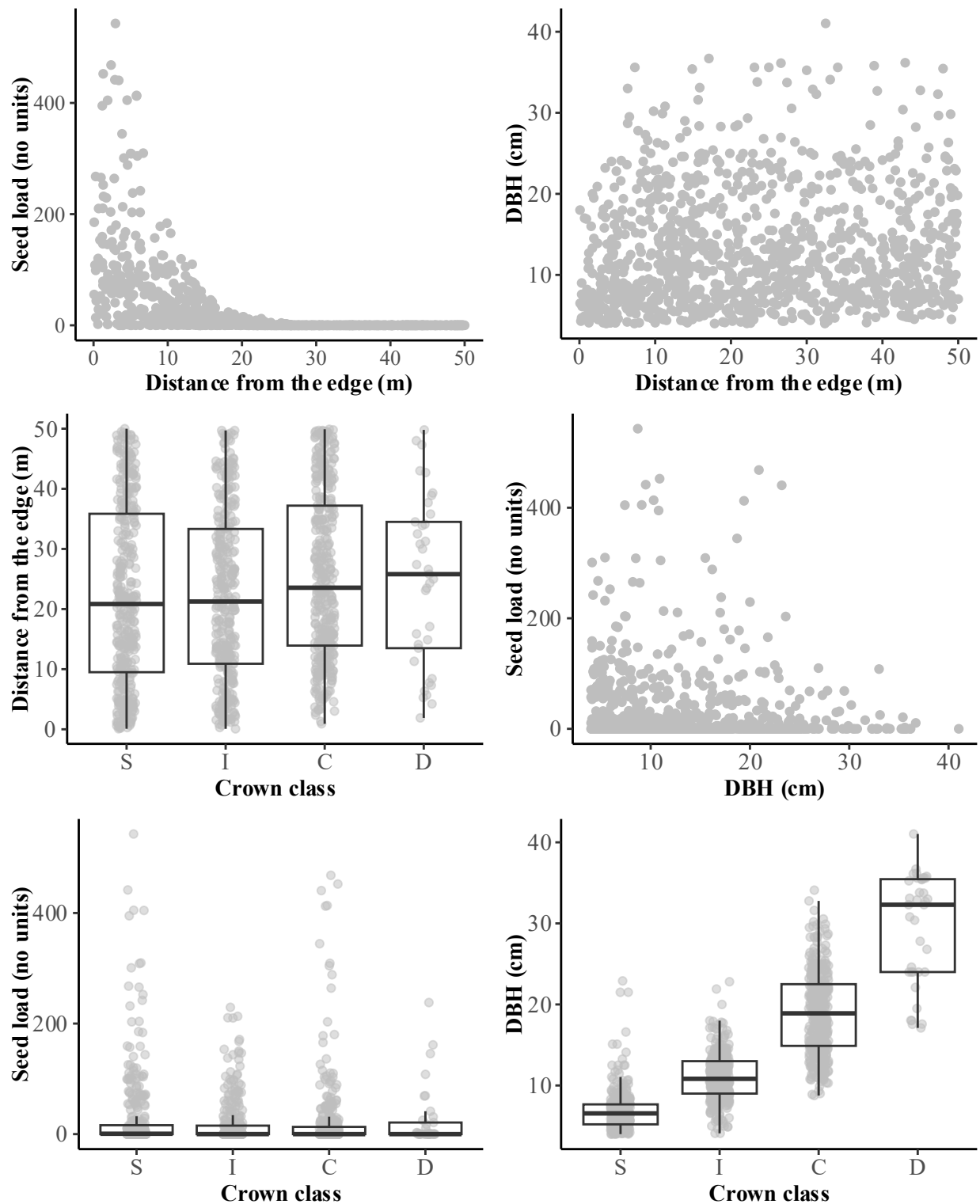


Figure A8. Pairs plot to assess collinearity between tree-level predictor variables. Data are the modelling dataset—live hemlock trees in the regenerating forest component from 0.0–50.0 m from the edge. Crown class codes: S = suppressed, I = intermediate, C = codominant, D = dominant. In boxplots, upper and lower edges of boxes are 75th and 25th percentiles; points beyond whiskers are outliers.

Table A5. Model fit statistics for full suite of variable combinations tested during model building.

Selected model is highlighted in yellow. Interactions terms are identified as: variable1*variable2.

Subscripts: (s) = scaled version of a variable, (n) = unique coefficient estimated for the variable for each logit (nominal effects, creating partial proportional odds model). Variables: DMR = dwarf mistletoe rating collapsed to three levels (0, IBLC–2, 3–5), DIST = horizontal distance from the mature-regenerating forest interface (m), SL = seed load (unitless), DBH = diameter at breast height (cm), CC = crown class, SITE = site id.

Model	Formula	Log-likelihood	AIC
m1	DMR ~ 1 (null)	-667	1338
Single variable models			
m2	DMR ~ DIST	-527	1061
m3	DMR ~ DIST ²	-527	1061
m4	DMR ~ DIST ⁴	-527	1061
m5	DMR ~ SL	-583	1172
m6	DMR ~ DBH	-653	1312
m7	DMR ~ CC	-652	1315
Two variable models			
m8	DMR ~ DIST + DBH	-482	972
m9	DMR ~ DIST + SL	-514	1037
m10	DMR ~ SL + DBH	-559	1126
Three variable model			
m11	DMR ~ DIST + SL + DBH	-469	948
Three variable models with first order interactions			
m12	DMR ~ DIST + SL + DBH + DIST*SL	-455	923
m13	DMR ~ DIST + SL + DBH + DIST*DBH	-466	943
m14	DMR ~ DIST + SL + DBH + SL*DBH	-460	932
m15	DMR ~ DIST + SL + DBH + DIST*SL + SL*DBH	-451	916
m16	DMR ~ DIST + SL + DBH + DIST*SL + DIST*DBH	-455	923
m17	DMR ~ DIST + SL + DBH + SL*DBH + DIST*DBH	-460	934
m18	DMR ~ DIST + SL + DBH + DIST*SL + DIST*DBH + SL*DBH	-451	918
Three variable model with first order interactions and random site effect			
m19	DMR ~ DIST + SL + DBH + DIST*SL + DIST*DBH + SL*DBH + (1 SITE)	-331	679
m20	DMR ~ DIST _s + SL _s + DBH _s + DIST _s *SL _s + SL _s *DBH _s + (1 SITE)	-331	679
Three variable model with first order interactions, random effect and nominal effects			
m21	same as model m20, fit with clmm2() function to allow comparisons with m22–26	-331	679
m22	DMR ~ DIST _{s,n} + SL _s + DBH _s + DIST _s *SL _s + SL _s *DBH _s + (1 SITE)	-328	673
m23	DMR ~ DIST _s + SL _{s,n} + DBH _s + DIST _s *SL _s + SL _s *DBH _s + (1 SITE)	-331	680
m24	DMR ~ DIST _s + SL _s + DBH _{s,n} + DIST _s *SL _s + SL _s *DBH _s + (1 SITE)	-331	680
m25	DMR ~ DIST _s + SL _s + DBH _s + (DIST _s *SL _s) _n + SL _s *DBH _s + (1 SITE)	-330	679
m26	DMR ~ DIST _s + SL _s + DBH _s + DIST _s *SL _s + (SL _s *DBH _s) _n + (1 SITE)	-331	680

Table A6. Likelihood ratio test results of variable significance during model building. Interactions terms are identified as: variable1*variable2. Subscripts: (s) = scaled version of a variable, (n) = unique coefficient estimated for the variable for each logit (nominal effects, creating partial proportional odds model). Variables: DIST = horizontal distance from the mature-regenerating forest interface (m), SL = seed load (unitless), DBH = diameter at breast height (cm), CC = crown class, SITE = site id. dAIC = difference in AIC, LR statistic = likelihood ratio test statistic. m20 and m21 are identical but fit with different functions. m20/m21 = the underlying model represented by m20 and m21.

Comparison	Variable tested	dAIC	LR statistic	df	p-value
Variable significance against null model					
m2-m1	DIST	-277	279	1	<0.0001
m3-m1	DIST ²	-277	279	1	<0.0001
m4-m1	DIST ⁴	-277	279	1	<0.0001
m5-m1	SL	-167	169	1	<0.0001
m6-m1	DBH	-26	27.7	1	<0.0001
m7-m1	CC	-23	29.4	1	<0.0001
Variable significance against single variable model					
m8-m2	DBH	-88	90.3	1	<0.0001
m8-m6	DIST	-199	342	1	<0.0001
m9-m2	SL	-24	26.1	1	<0.0001
m9-m5	DIST	-135	137	1	<0.0001
m10-m5	DBH	-46	47.9	1	<0.0001
m10-m6	SL	-187	189	1	<0.0001
Variable significance against two variable model					
m11-m8	SL	-25	26.5	1	<0.0001
m11-m9	DBH	-89	90.7	1	<0.0001
m11-m10	DIST	-178	180.	1	<0.0001
Interaction term significance against three variable model					
m12-m11	DIST*SL	-25	27.0	1	<0.0001
m13-m11	DIST*DBH	-5	6.70	1	0.0094
m14-m11	SL*DBH	-16	17.9	1	<0.0001
Interaction term significance against three variable model with one interaction term					
m15-m12	SL*DBH	-7	9.00	1	0.0027
m15-m14	DIST*SL	-16	18.1	1	<0.0001
m16-m12	DIST*DBH	0	1.90	1	0.164
m16-m13	DIST*SL	-20	22.2	1	<0.0001
m17-m13	SL*DBH	-9	11.3	1	0.0008
m17-m14	DIST*DBH	2	0.20	1	0.694
Interaction term significance against three variable model with two interaction terms					
m18-m15	DIST*DBH	2	0.00	1	0.931
m18-m16	SL*DBH	-5	7.10	1	0.0079
m18-m17	DIST*SL	-16	17.9	1	<0.0001
Random effect significance					
m19-m15	(1 SITE)	-237	239	1	<0.0001
Nominal effect significance (= proportional odds assumption test)					
m22-m20/m21	DIST _{s,n}	-5	7.40	1	0.0067
m23-m20/m21	SL _{s,n}	1	1.00	1	0.314
m24-m20/m21	DBH _{s,n}	2	0.200	1	0.646
m25-m20/m21	(DIST _s *SL _s) _n	0	1.70	1	0.196
m26-m20/m21	(SL _s *DBH _s) _n	1	0.800	1	0.378

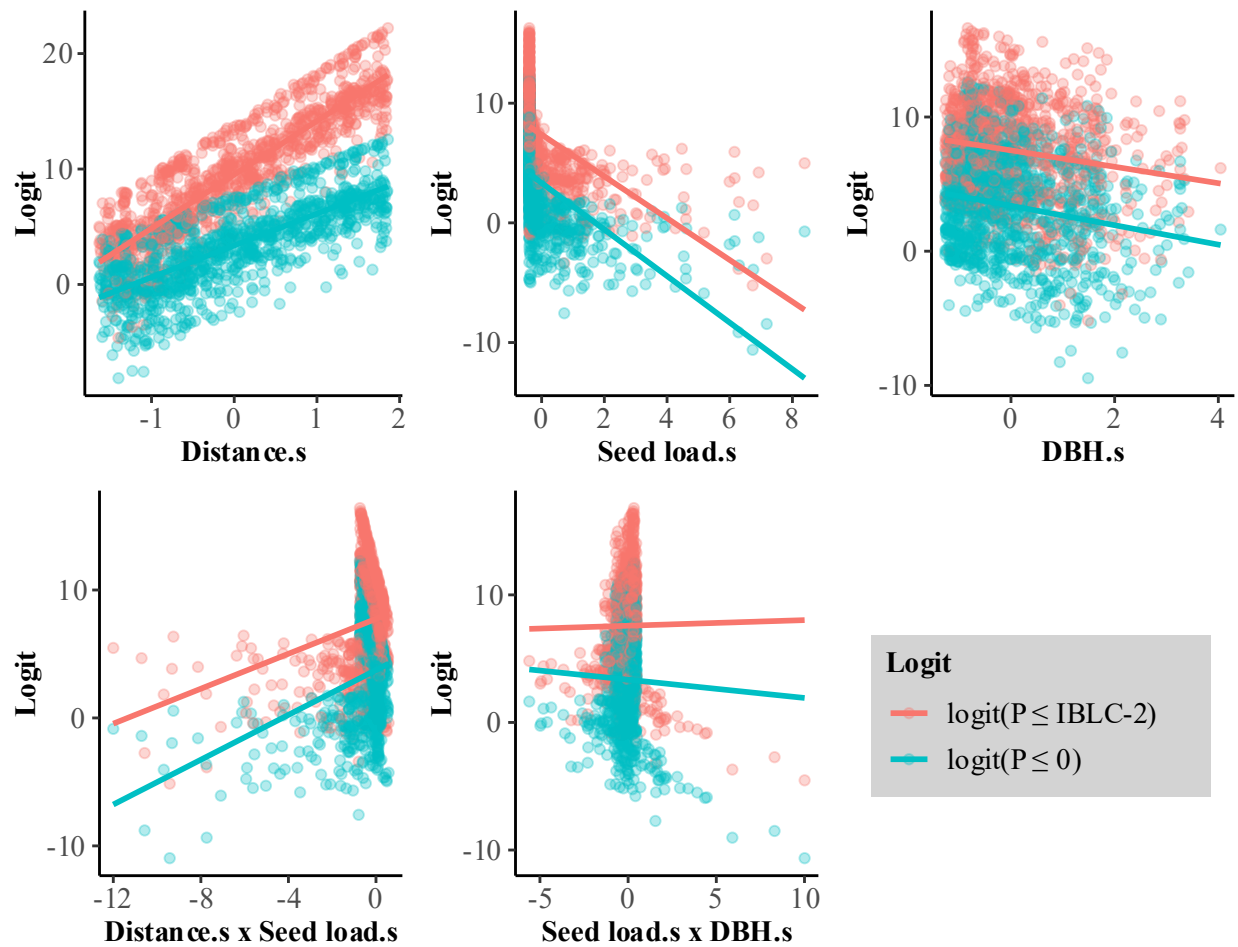


Figure A9. Graphical test of proportional odds assumption for each predictor variable. Each plot shows the logit odds of the response (dwarf mistletoe rating (DMR) collapsed to three classes (0, IBLC–2, 3–5)) being $\leq$ a given class. logit predictions come from models that allow for unique coefficients for each logit for a given focal variable (m22–m26 in Tables A5 and A6). Data are the observations in the modelling dataset—live hemlock (measured and simulated) in the regenerating component within 50m of the edge. Assumption is met if logit odds (lines) of the response levels are parallel across range of variable. “.s” suffix indicates variables are scaled.

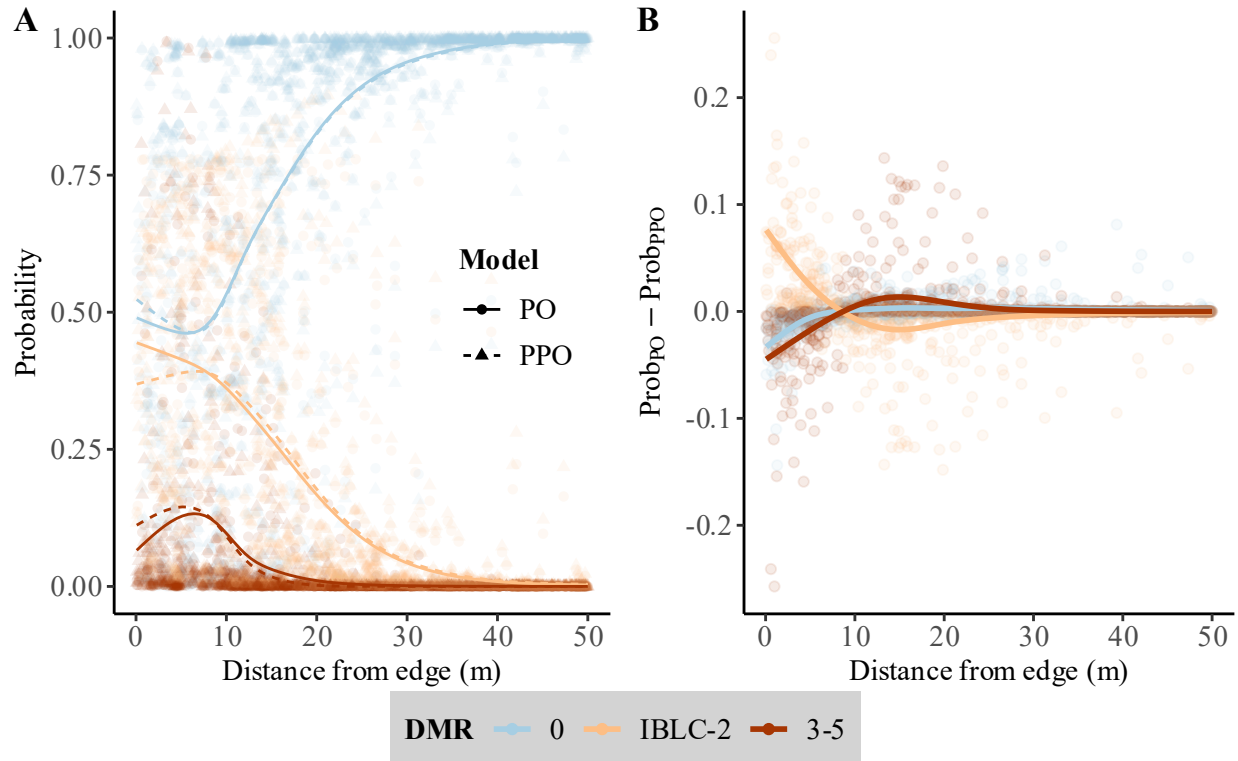


Figure A10. Predicted probabilities of each dwarf mistletoe rating (DMR) class from models with (PO) and without (PPO) proportional odds assumption on distance from the edge variable. PO model is m20/21 in Tables A5 and A6. PPO model is m22 in Tables A5 and A6. (A) Raw probabilities for each model. (B) Difference in probabilities between the models. Divergence of lines in (A) and magnitude of difference in (B) indicate level of model disagreement across the range of the distance from the edge variable. Data are observations in the modelling dataset—live hemlock (measured and simulated) in the regenerating component within 50.0 m of the edge. Trend lines are generalized additive models fit to the values. IBLC = infected below live crown.

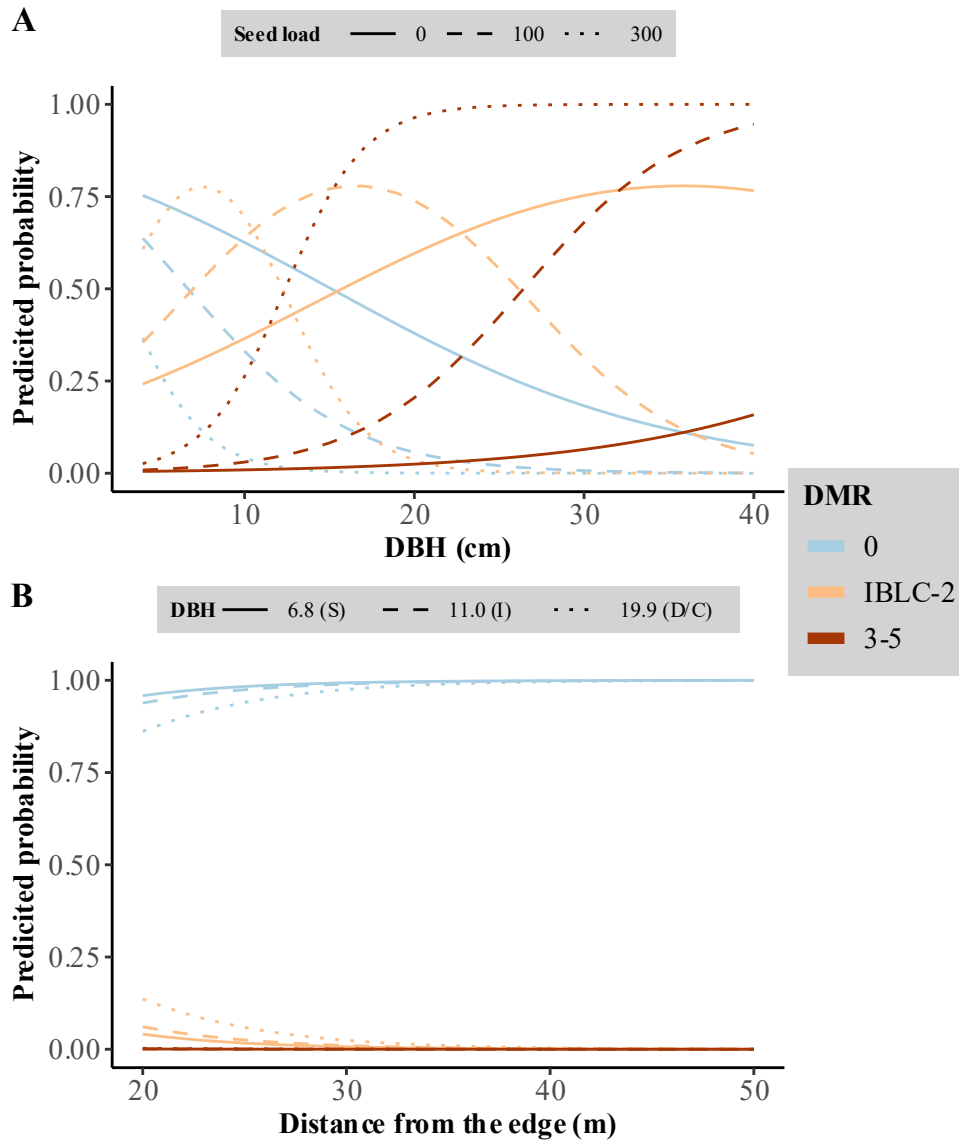


Figure A11. (A) Effect of DBH on predicted probabilities of dwarf mistletoe rating (DMR) classes for trees at 7.5 m from the edge. (B) Comparison of DBH and distance from the edge effects on predicted probabilities of DMR classes for trees 20.0 to 50.0 m from the edge. Near the edge (as in A), the relative probabilities of each DMR class depend strongly on DBH, but this relationship is affected by seed load. Beyond 20.0 m (as in B), seed load is ~0 and the effect of DBH is small relative to the effect of increasing distance from the edge.

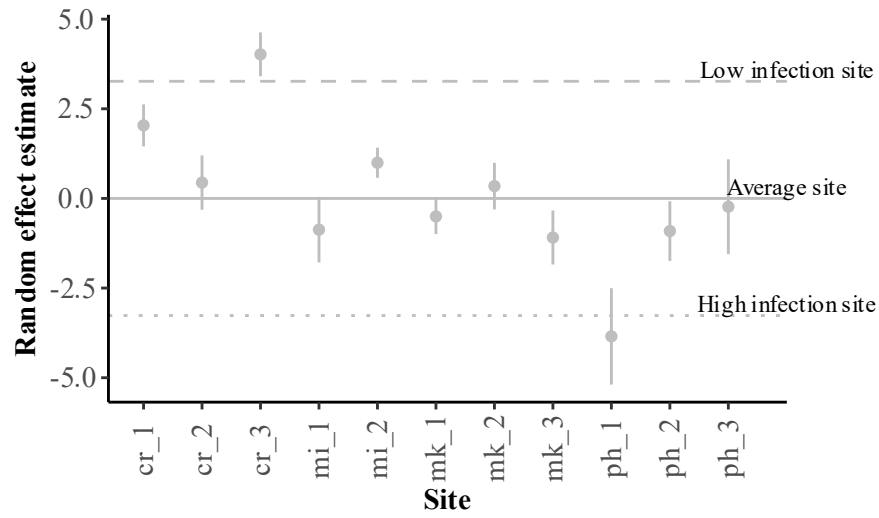


Figure A12. Random site effect estimates for each site. Estimates are the conditional modes with 95% confidence intervals based on conditional variances. The estimates are used to define hypothetical average infection (random effect = 0), high infection (upper 95th percentile) and low infection sites (lower 95th percentile) sites. These hypothetical sites are used to demonstrate the magnitude of the random site effect on predicted probabilities of DMR infection classes in Figure 11 in main text.

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